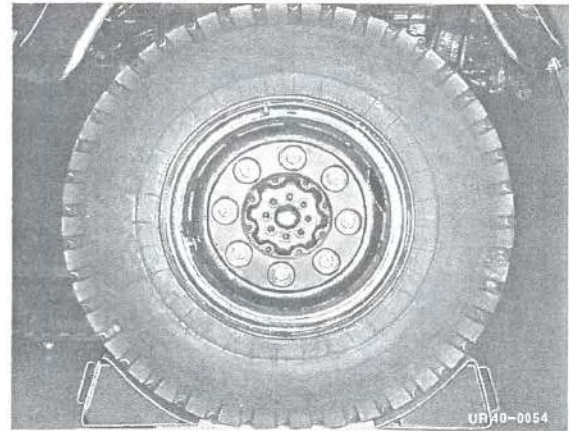


Disassembly

Note: Operations at left and right-hand wheel hub drives are to be executed in same order. Scope of work is same with axle in situ.

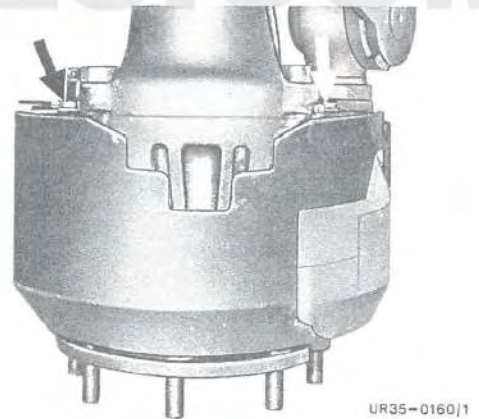
- 1 Remove rear axle.
- 2 Detach wheels.



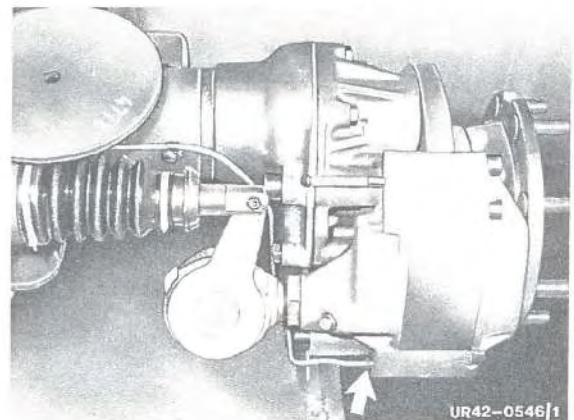
- 3 Drain oil off wheel hub drive.

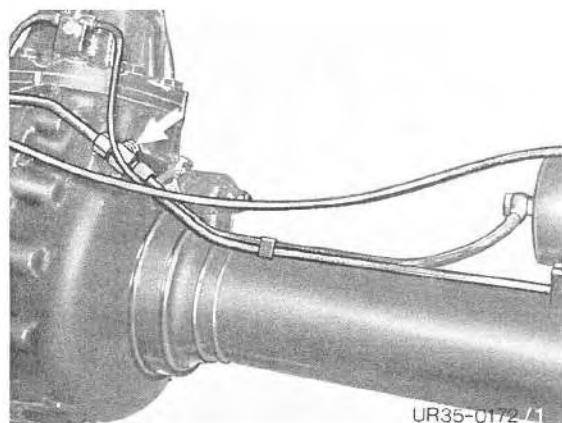


- 4 Remove brake backplate.

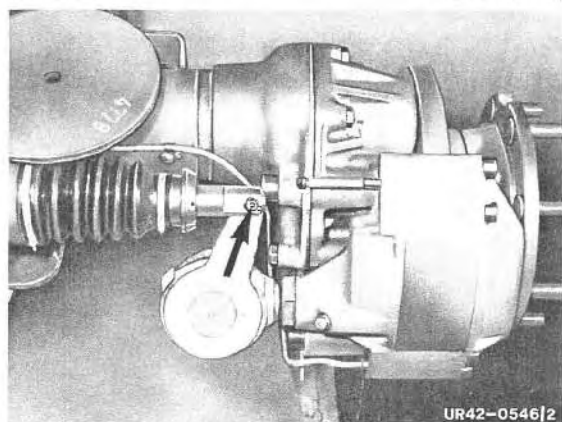


- 5 Unscrew brake line at fixed caliper and seal end.

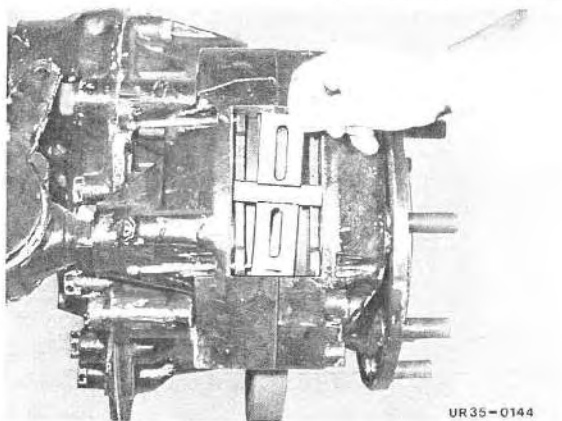




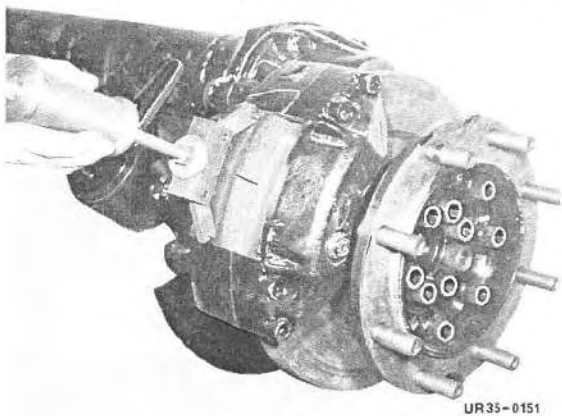
6 Admit at least 6 bar to spring-loaded cylinder via junction in order to release brakes and actuate quick release.



7 Release and remove pin in clevis of spring-loaded cylinder.

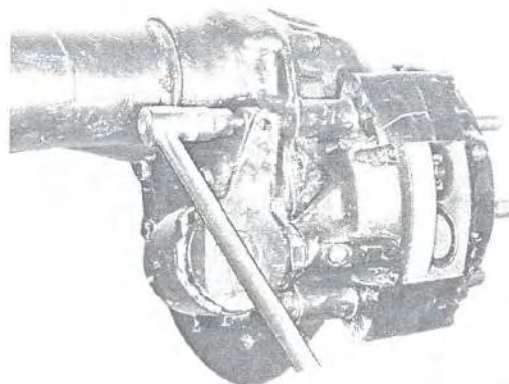


8 Detach metal cover at fixed caliper, knock pins out and remove spring clip.



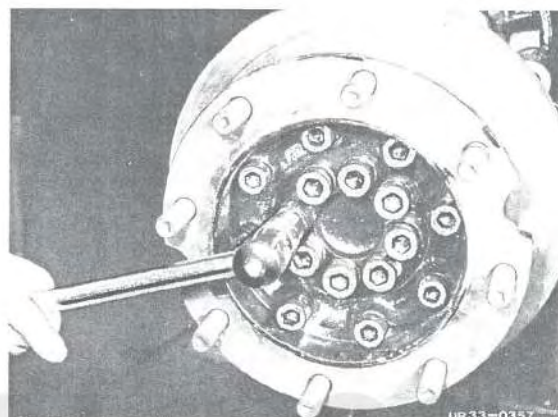
9 Remove brake pads using special tool No. 10.

- 10 Unscrew fixed caliper using special tool No. 3.



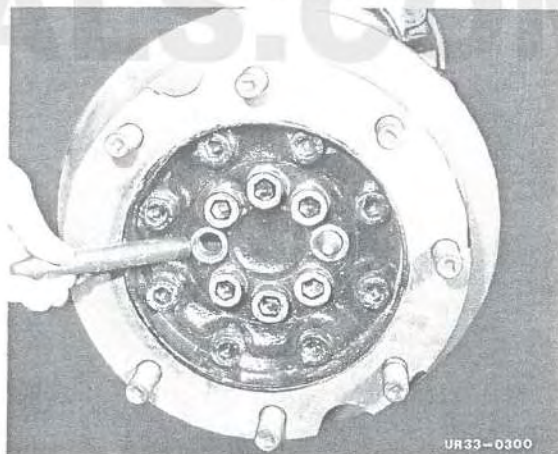
UR 35-0150

- 11 Unscrew two wheel hub fastening bolts using special tool No. 3.



UR 33-0357

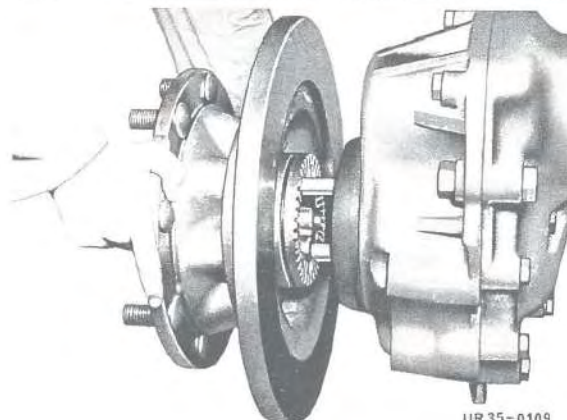
- 12 Insert special tool No. 6 and unscrew all other fastening bolts.



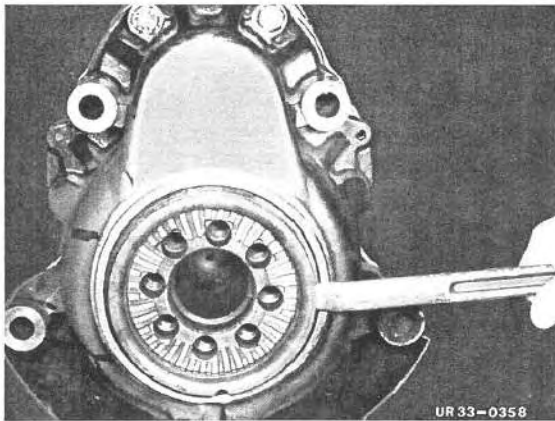
UR 33-0300

- 13 Remove wheel hub with brake disk.

- 14 Remove oil baffle.



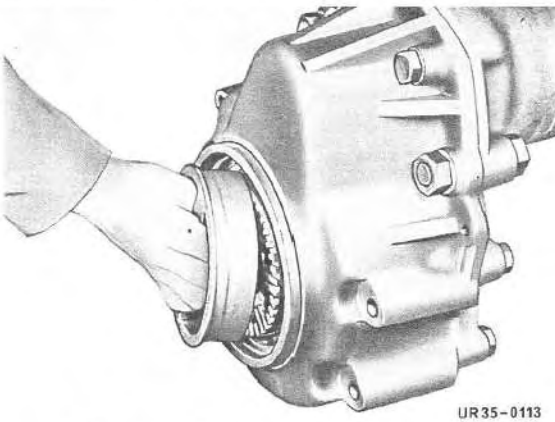
UR 35-0109



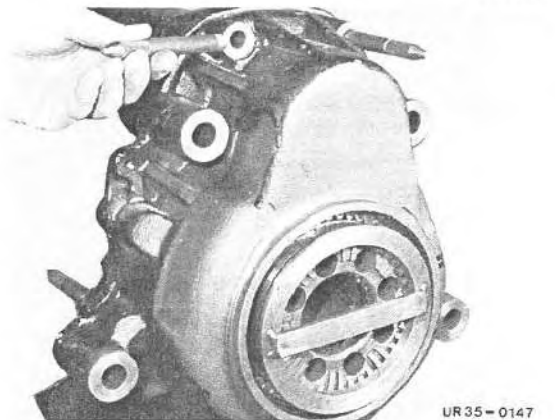
15 Remove wear ring.

Note: Don't tilt wear ring during removal for to avoid damage.

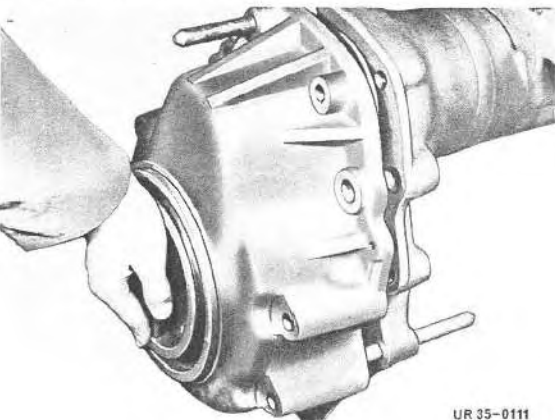
16 Remove shaft seal.



17 Introduce special tool No. 4 into roller bearing.



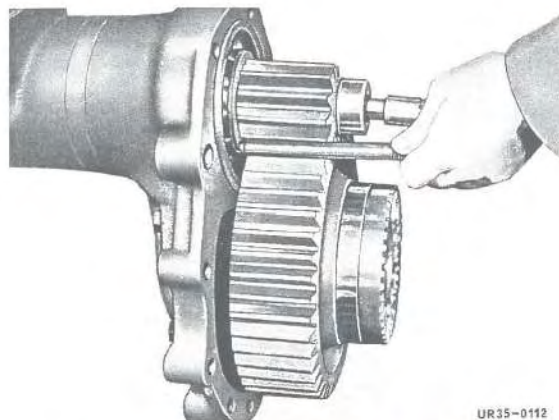
18 Release wheel hub drive housing and insert special tool No. 6.



19 Remove housing, simultaneously slipping special tool No. 4 fully home into roller bearing so that rollers cannot drop out.

Note: Special tool No. 4 is to be held by O-ring so that it cannot drop out of bearing.

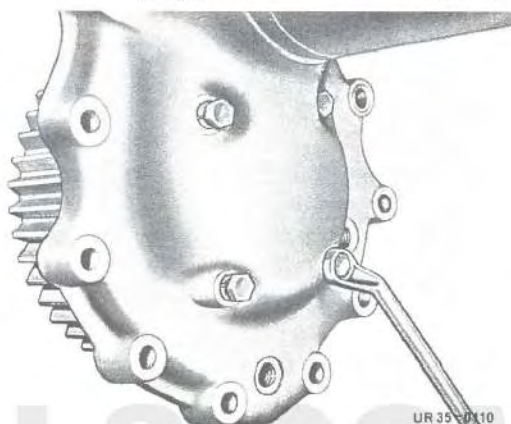
20 Lock drive gear to driven gear (e.g. by inserting a copper drift). Unscrew center fastening bolt using special tool No. 3.



UR35-0112

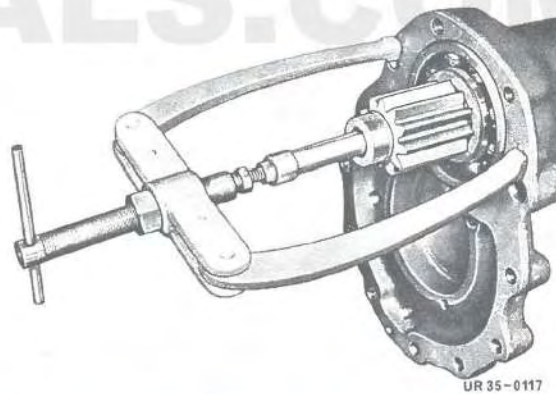
21 Release fastening bolts of bearing ring at axle tube flange. Withdraw driven gear using special tools No. 8 and No. 9.

Caution — danger: Attach special tool to crane because driven gear may otherwise drop.



UR35-0110

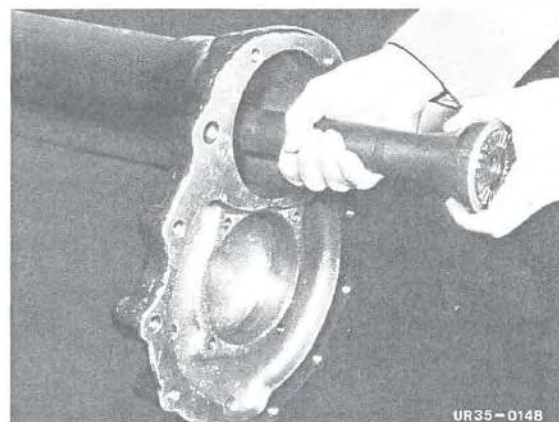
22 Withdraw drive gear using puller and special tool No. 9.



UR 35-0117

23 Remove rear axle shaft from axle tube.

24 Clean and check all parts, exchange if necessary.



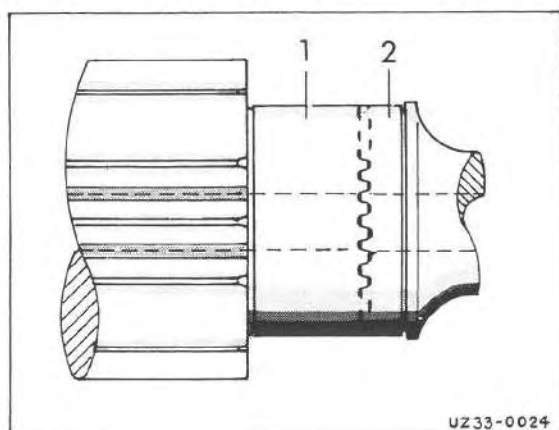
UR35-0148

747.2

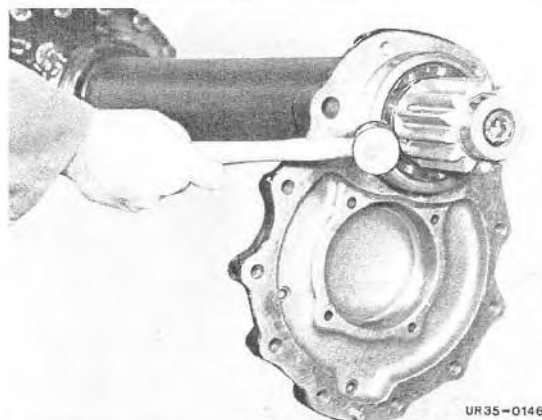


Assembly

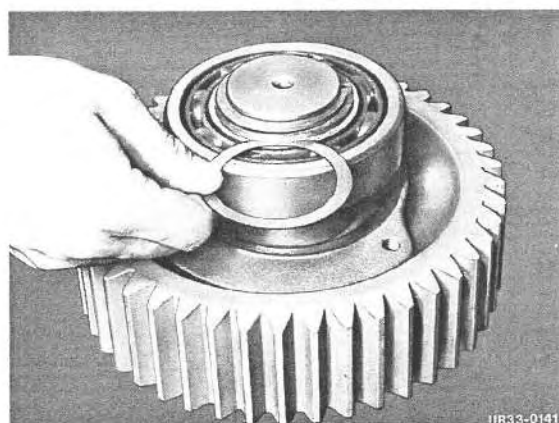
- 1 Press deep-groove ball bearing onto drive gear.



- 2 Clamp rear axle shaft in vise and bolt drive gear (1) to rear axle shaft (2) so that teeth mesh.

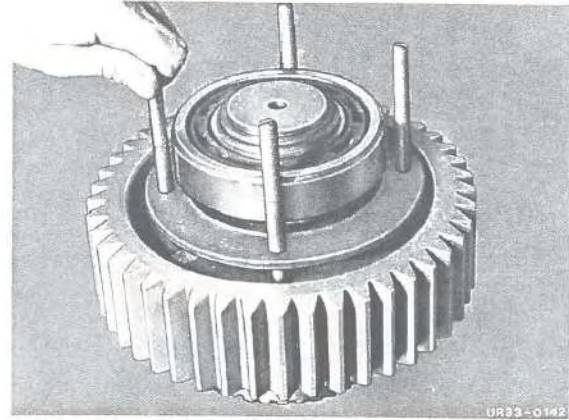


- 3 Install assembled rear axle shaft in differential and, using plastic-headed hammer, drive deep-groove ball bearing of drive gear into bearing seat.



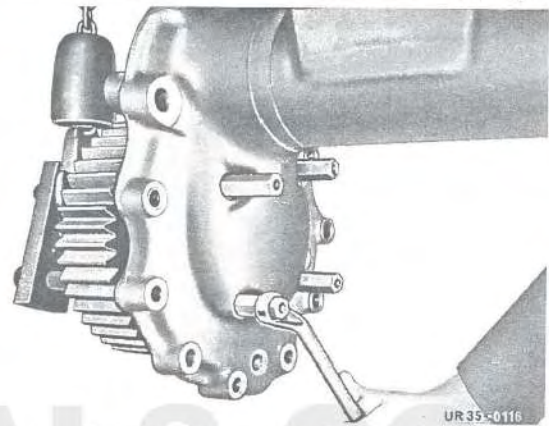
- 4 If deep-groove ball bearing at driven gear is worn, insert bearing ring and force deep-groove ball bearing into position.
- 5 Fit supporting ring and snap ring. Obtain distance "0" between supporting ring and driven gear by using suitable shim.

- 6 Screw special tool No. 8 into bearing ring. Introduce driven gear into axle tube flange.



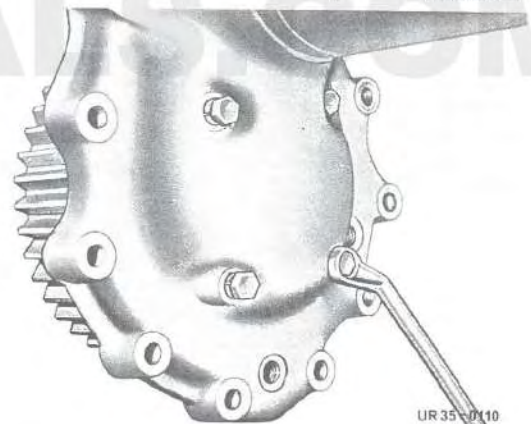
- 7 Using special tool No. 8, draw driven gear evenly into bearing seat.

Note: Make sure that contact flange of bearing ring points toward deep-groove ball bearing of drive gear.

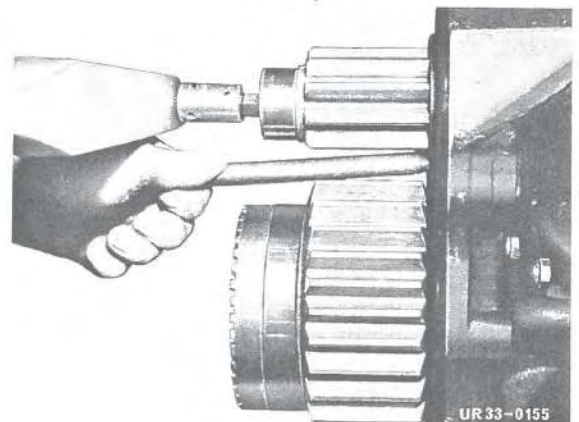


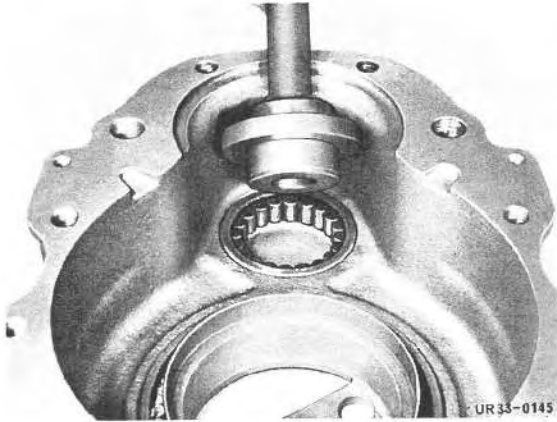
Note:

- 8 Coat fastenings bolts with sealant No. 2 and tighten driven gear.

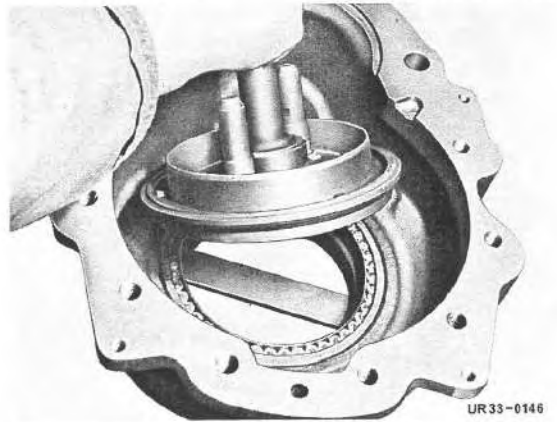


- 9 Lock drive gear to driven gear (e.g. by inserting a copper drift). Using special tool No. 3, torque center fastening bolt. See 1.5/1 for tightening torques.

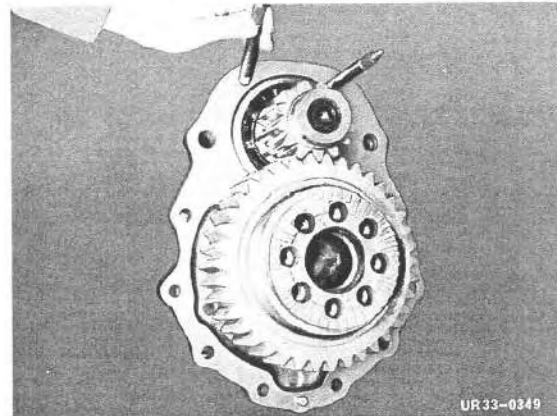




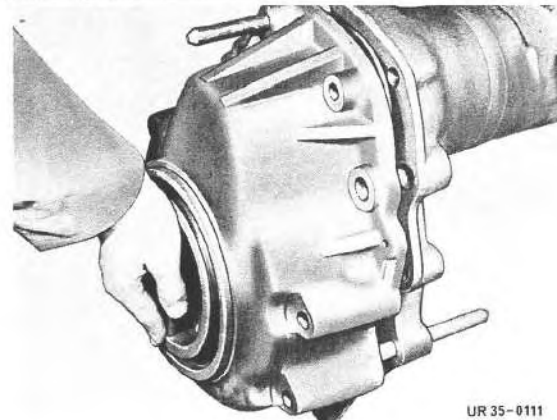
10 If small roller bearing in wheel hub drive housing is defective, press new roller bearing into wheel hub drive housing with special tool No. 19.



11 In order to install large roller bearing for driven gear, substitute special tool No. 4 for transport sleeve. Force roller bearing into position using special tool No. 12.



12 Screw special tool No. 6 into axle tube flange.



13 Coat contact surface of wheel hub drive housing with sealant No. 3 and attach wheel hub drive housing.

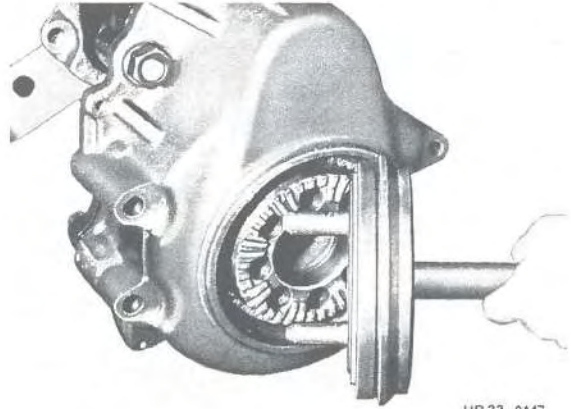
Note: Watch out for special tool No. 4 so that no rollers drop out.

14 Coat shaft seal with grease No. 2 and install using special tool No. 12.

Note:

From model	Chassis end No.	Rear axle No.
435.111	071 365	

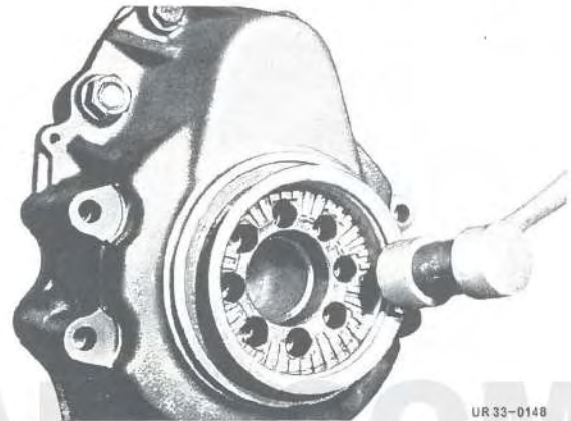
New shaft seal installed with modified lips and made of new material for improved sealing properties.



UR 33-0147

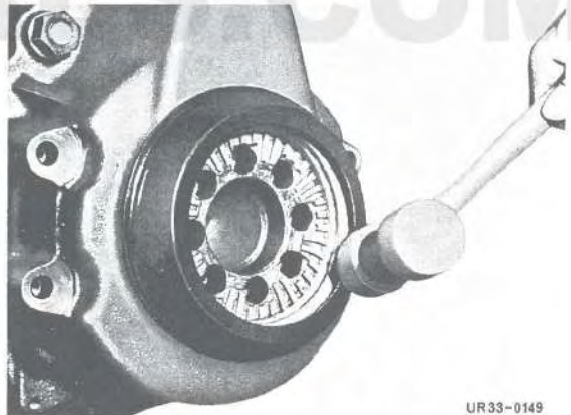
15 Install wear ring.

Note: If dust lip of shaft seal is drawn in by wear ring, pull out dust lip using piece of wire.



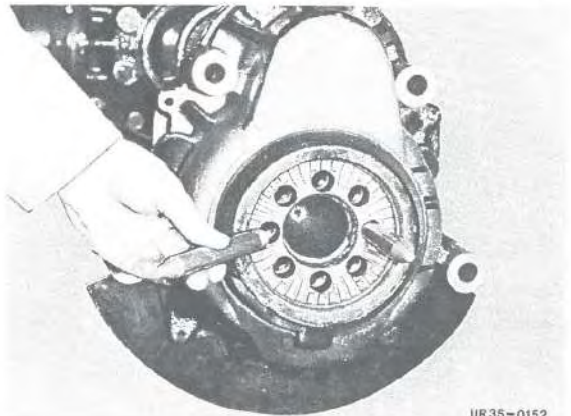
UR 33-0148

16 Attach oil baffle.



UR33-0149

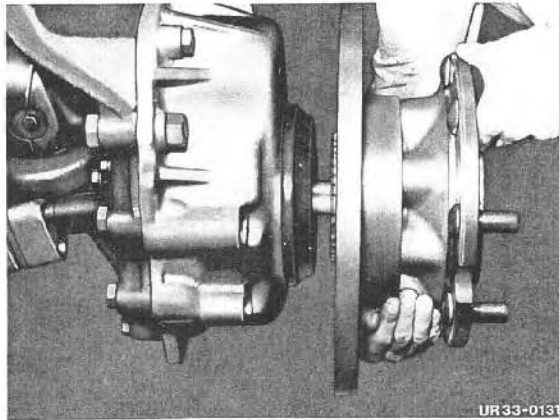
17 Screw special tool No. 5 to output gear.



UR 35-0152

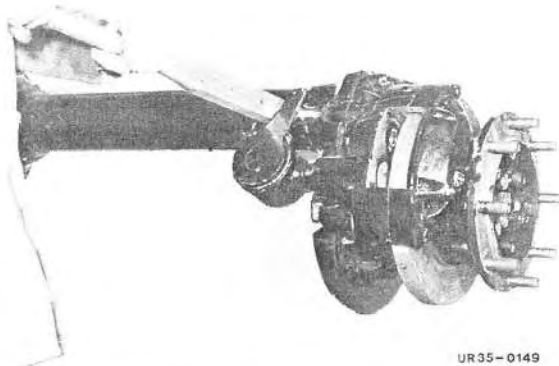
35.3 Disassembly and Assembly of Wheel Hub Drive

747.2



18 Fit O-ring on wheel hub.

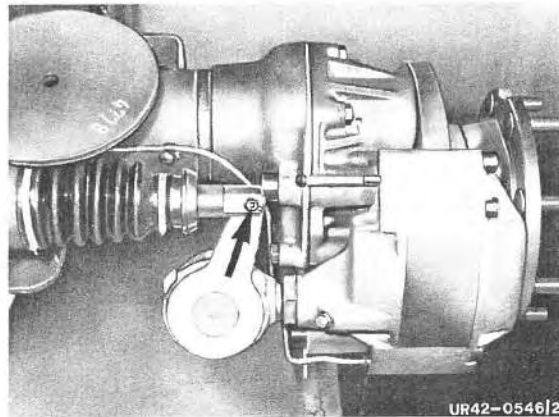
19 Slide wheel hub and brake disk over special tool No. 5 and tighten.
Tightening torques, see 1.5/1.



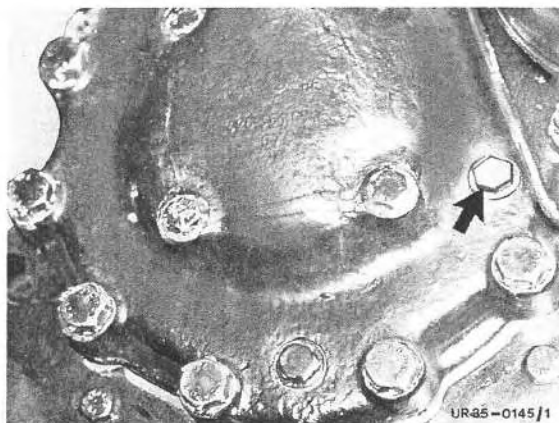
20 Tighten fixed caliper using special tool No. 3.
Tightening torques, see 1.5/1.

21 Connect brake line to fixed caliper.

22 Install brake pads.



23 Coat split pin in clevis of spring-loaded cylinder with paste No. 6, install and secure. Check setting of quick release device and adjust if necessary.
Setting, see 1.3/1.



24 Fill wheel hub drive with oil.
Capacity, see 1.2/1.

25 Attach brake backplate.

26 Install rear axle.

27 Fit wheels and tighten wheel nuts.
Tightening torques, see 1.5/1.

Note: Retighten wheel nuts after 50 km.

28 Bleed brake system and check for leaks.

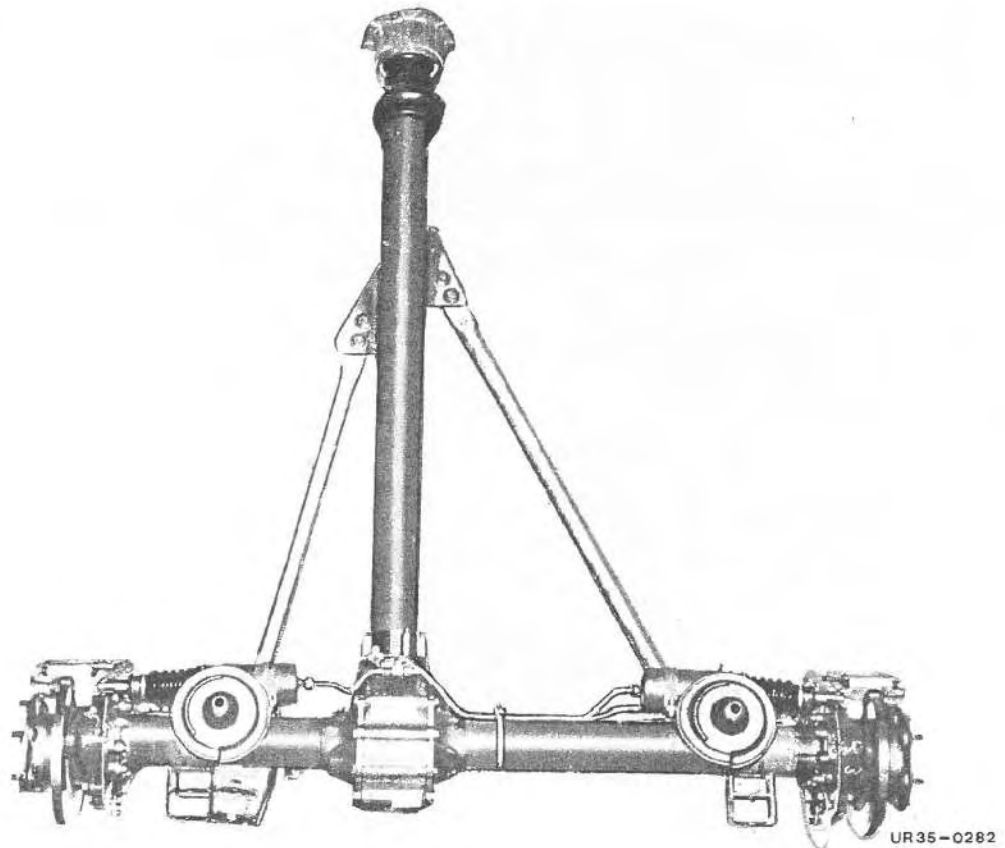
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Worksheet for Adjusting Axle Drive	1.7/1
2 Removal and Installation of Rear Axle	2.1/1
3 Disassembly and Assembly of Rear Axle	3.1/1
4 Disassembly and Assembly of Drive Pinion Bearing	4.1/1
5 Disassembly and Assembly of Differential Gearing	5.1/1
6 Disassembly and Assembly of Wheel Hub Drive	6.1/1
7 Checking Differential Lock for Leaks	7.1/1
8 Renewing Torque Tube Sleeve	8.1/1

Installation Survey

Chassis		Rear axle		Brake	Installation
Model	Sales designation	Model	Sales designation		
435.115	U 1300 L	747.111	HU 2/14 S - 4,0	Disk brake	Series
435.117	U 1300 L/37	747.113			

Overall View



Rear axle

35.6 General

747.111/1113

Technical Data

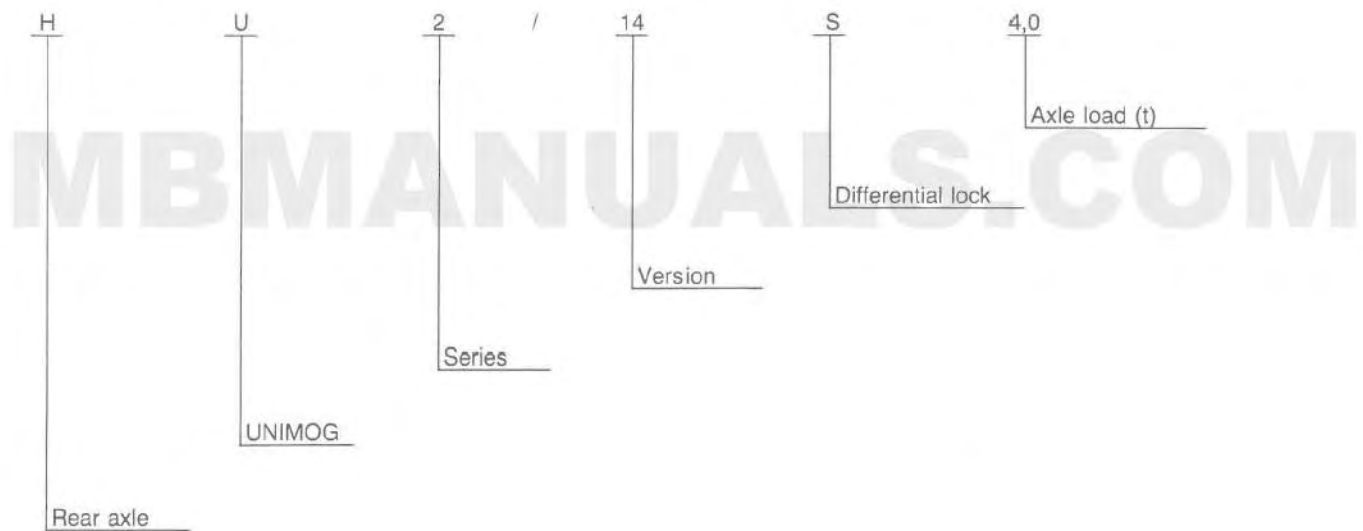
Ratios

Axle model	Installation	Gear train Ring gear drive pinion		Gear train Wheel hub drive		Total Ratio
		Number of teeth	Ratio	Number of teeth	Ratio	
747.111 .113	Series	22: 7	3,143	27: 13	2,077	6,527
747.111 .113	SA	23: 9	2,555			5,307

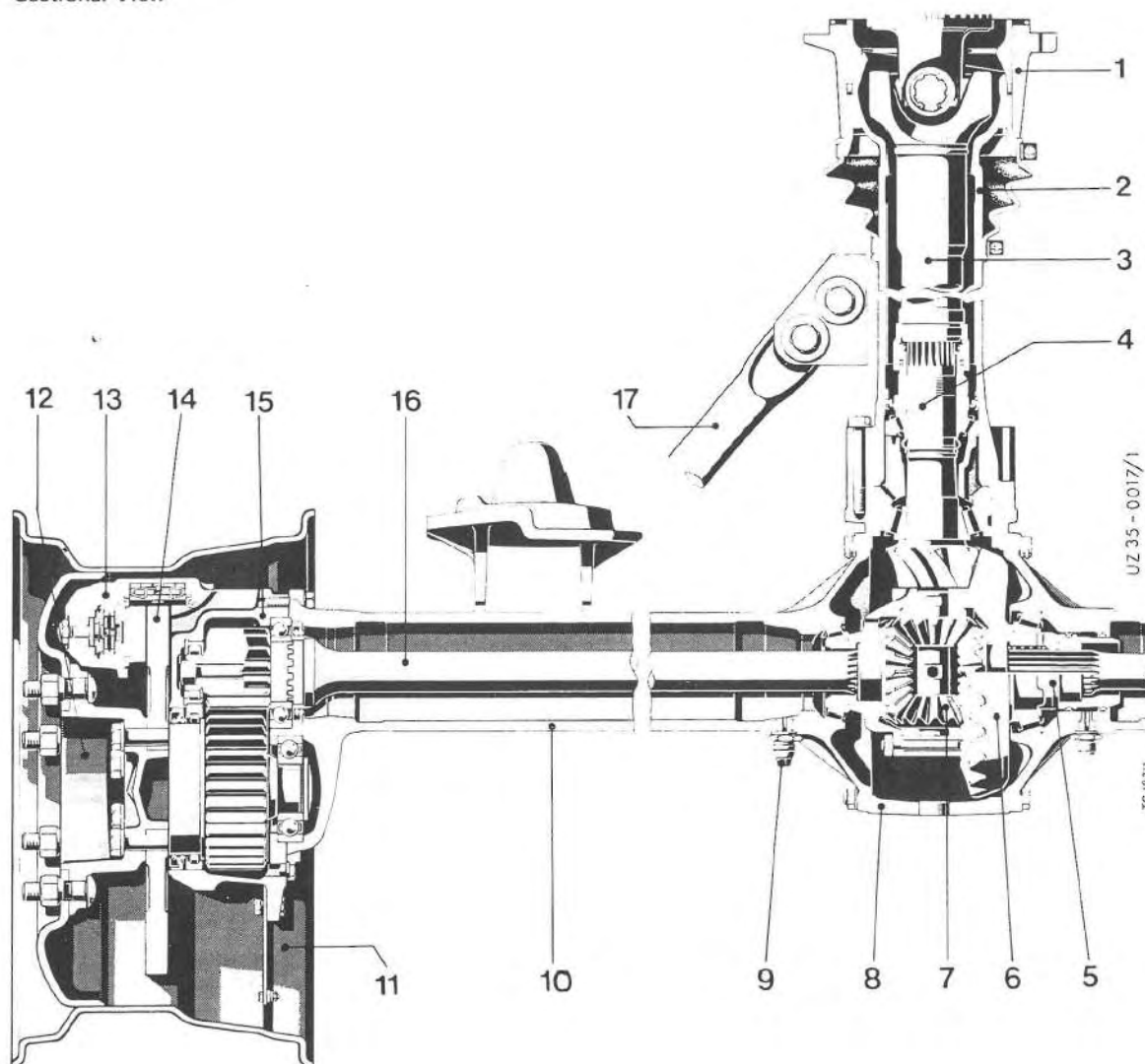
Capacity

Designation	Service product	SAE class	Capacity (L)	Axle model
Axle center housing	Hypoid gear oil	90	2,5	747.111
Wheel hub drive			each 0,3	.113

Key to Rear Axle Designation



Sectional View

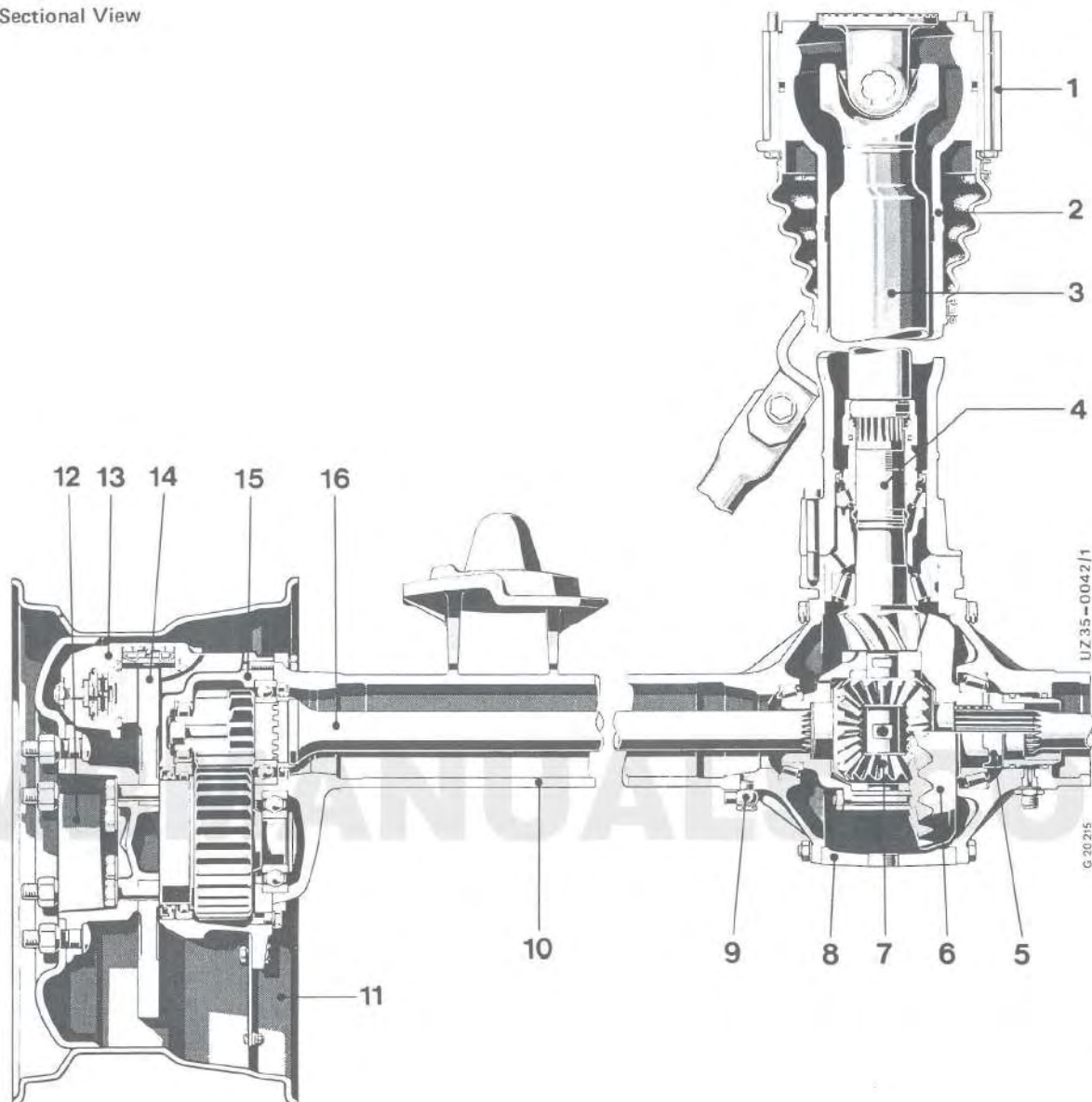


HU 2/14S-4.0

- | | |
|----------------------------------|--------------------|
| 1 Torque ball casing with shells | 10 Axle tube |
| 2 Torque tube | 11 Disk wheel |
| 3 Drive shaft | 12 Wheel hub |
| 4 Drive pinion | 13 Fixed caliper |
| 5 Differential lock | 14 Brake disk |
| 6 Ring gear | 15 Wheel hub drive |
| 7 Differential | 16 Rear axle shaft |
| 8 Center axle housing | 17 Axle strut |
| 9 Axle ventilation connection | |

747.113

Sectional View



HU 2/14 S-4.0

- | | |
|----------------------------------|--------------------|
| 1 Torque ball casing with shells | 10 Axle tube |
| 2 Torque tube | 11 Disk wheel |
| 3 Drive shaft | 12 Wheel hub |
| 4 Drive pinion | 13 Fixed caliper |
| 5 Differential lock | 14 Brake disk |
| 6 Ring gear | 15 Wheel hub drive |
| 7 Differential | 16 Rear axle shaft |
| 8 Center axle housing | 17 Axle strut |
| 9 Axle ventilation connection | |

Setting Values

Designation	Setting Value
Preload the torque ball shells on extension tube mounting flange manual effort measured	200 to 300 N
Breakaway force for drive pinion, refer to axle data sheet	3,0 to 3,5 Nm
Backlash in differential	0,15 to 0,20 mm
Claw play of differential lock	0,5 $\pm$ 0,2 mm
End play bearing on output gear	0
Axial play of bearing in wheel hub drive	0 No preload
Bearing play of tapered roller bearing on differential	0 No preload
Adjustment between hole of actuating lever and clevis for spring brake	78 mm $\pm$ 0,5
Brake pad clearance	0,30 mm

Necessary Materials

Cons. No.	Designation	Part number
1	Teroson Atmosit sealing compound	000 989 43 20 ¹⁾
2	Longlife grease	commercially available
3	Molybdenum disulphide running-in paste	commercially available
4	Loctite No. 241 adhesive	002 989 70 71
5	Molykote liquid grease	commercially available
6	Anti-corrosive oil SAE 30	commercially available
7	Hypoid gear oil SAE 90	commercially available
8	Loctite No. 573	commercially available
9	Terostat 56 surface sealing compound	001 989 58 20

1) optional 001 989 58 20

General note: In order to ensure satisfactory adhesion of sealing compounds, the sealing surfaces must be completely free of oil and grease.

Special Tools

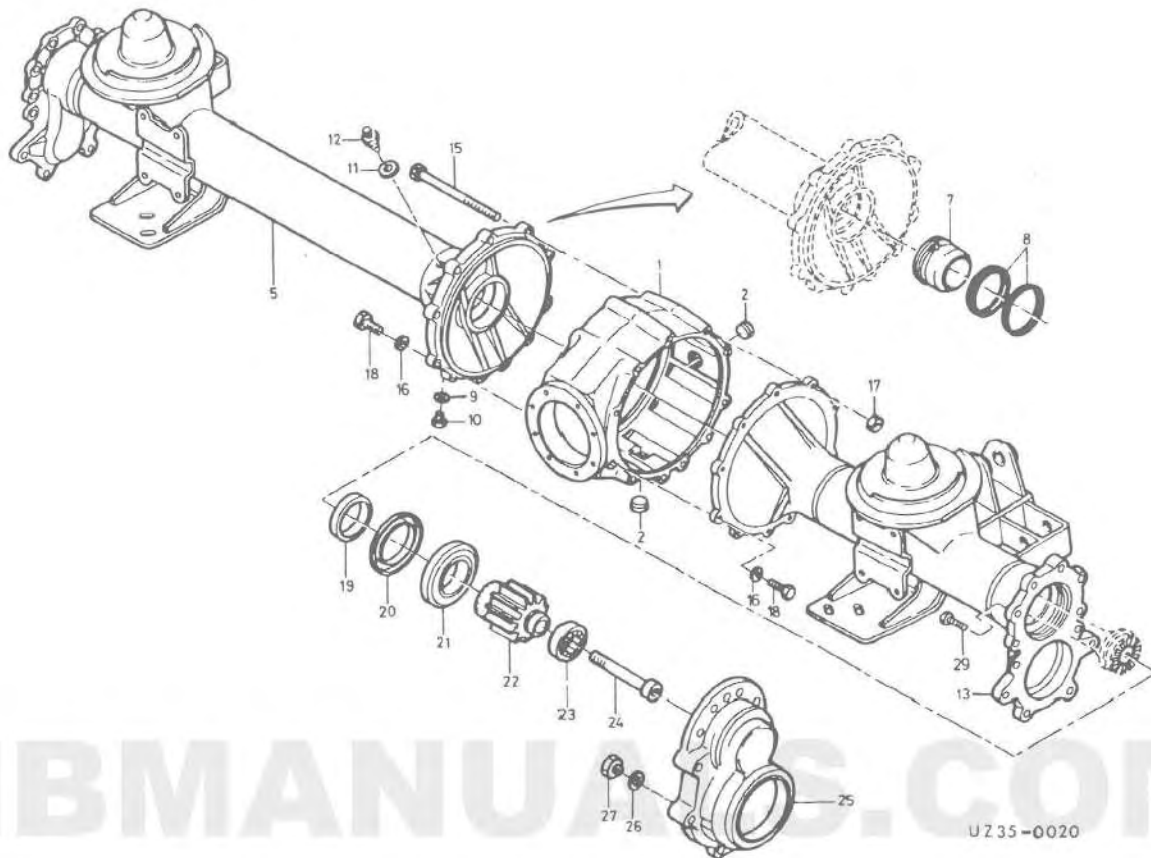
Cons. No.	Tool	Number	Tool set
1	Extension tube for mounting flange	000 589 01 14 00	A
2	Allan key socket 10 mm for torque tube and wheel hub drive	001 589 61 09 08	B
3	Allan key socket 14 mm, 3/4" square for drive gear	001 589 68 09 00	B
4	Allan key socket 24 mm for wheel hub	001 589 71 09 00	B
5	Assembly drift (two) for wheel hub and brake disk	424 589 00 15 00	B
6	Drift for installer	406 589 04 15 00	B
7	Locking fixture for drive gear	435 589 01 31 00	B
8	Counter-support for extractor	000 589 35 33 00	B
9	Extractor for brake pads	001 589 43 33 00	B
10	Installer for shaft seal in torque tube	435 589 01 35 00	B
11	Installer for shaft seal in steering knuckle and axle tube	424 589 00 43 00	B
12	Installer for roller bearing and shaft seal in wheel hub drive	425 589 01 43 00	B
13	Piston resetting fixture for brake piston	000 589 52 43 00	B
14	Assembly flange for torque tube (with tool No. 1)	425 589 00 59 00	B
15	Insertion sleeve for torque tube	406 589 01 59 00	B
16	Adjusting key for drive pinion	000 589 64 03 00	C
17	Adjuster for drive pinion	425 589 00 21 00	C
18	Torque wrench for drive pinion (0 to 5 Nm) 1/4" square	000 589 87 21 00	C
19	Pressure tester for differential lock (checking for leakage)	401 589 01 25 00	C
20	Mounting for drive pinion	425 589 00 31 00	C
21	Installer for taper roller bearings in axle tubes and differential (with tool No. 6)	406 589 02 43 00	C
22	Installer for taper roller bearings on differential (with tool No. 6)	406 589 03 43 00	C
23	Installer for small roller bearing in wheel hub drive	406 589 05 43 00	C

Tightening torques

Designation	Thread	Nm
Wheel hub to drive gear	M 16 x 1,5– 12.9	320 ± 20
Nut clamping bolt output gear	M 20 x 1,5– 10.9	220
Drive gear to drive shaft	M 16 x 1,5– 12.9	300
Fixed caliper to wheel hub drive housing	M 16 x 1,5– 10.9	250
Shock absorber mounting to axle	M 20 x 1,5– 8.8	250
Shock absorber mounting to frame	M 20 x 1,5– 8.8	250
Axle strut to axle tube	M 18 x 1,5– 10.9	350
Axle strut to torque tube	M 18 x 1,5– 10.9	350
Transverse link to frame	M 16 x 1,5– 10.9	315
Transverse link to axle	M 16 x 1,5– 10.9	315
Rear spring to frame top	M 18 x 1,5– 8.8	315
Axle tube assembly to axle center housing	M 14 x 1,5– 12.9	200
Ring gear to differential case	M 12 x 1,5	120
Torque tube to axle center housing	M 12 x 1,5– 10.9	90
Wheel nuts ¹⁾	A 18	400

1) For type 435.117 with rim 11 - 20 SDC (hub centering) 450 Nm.

Exploded view

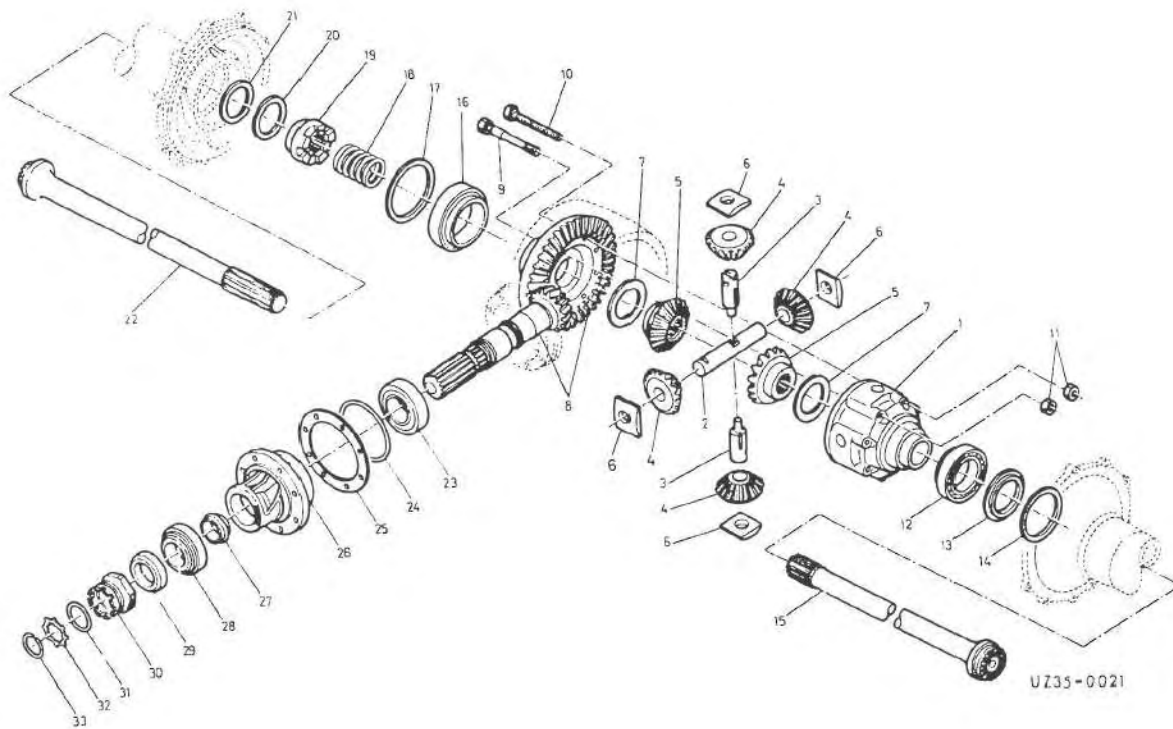


Axle tubes and center axle housing

- | | |
|-----------------------|-------------------------------|
| 1 Center axle housing | 17 Nut |
| 2 Screw plug | 18 Bolt |
| 5 Axle tube | 19 Wear ring |
| 7 Piston | 20 Sealing ring |
| 8 Sealing ring | 21 Ball bearing |
| 9 Sealing ring | 22 Drive gear |
| 10 Bolt | 23 Cylindrical roller bearing |
| 11 Sealing ring | 24 Bolt |
| 12 Union | 25 Housing |
| 13 Axle tube | 26 Sealing ring |
| 15 Bolt | 27 Screw plug |
| 16 Washer | |

Text and illustration do not correspond to microfiche.

Exploded view

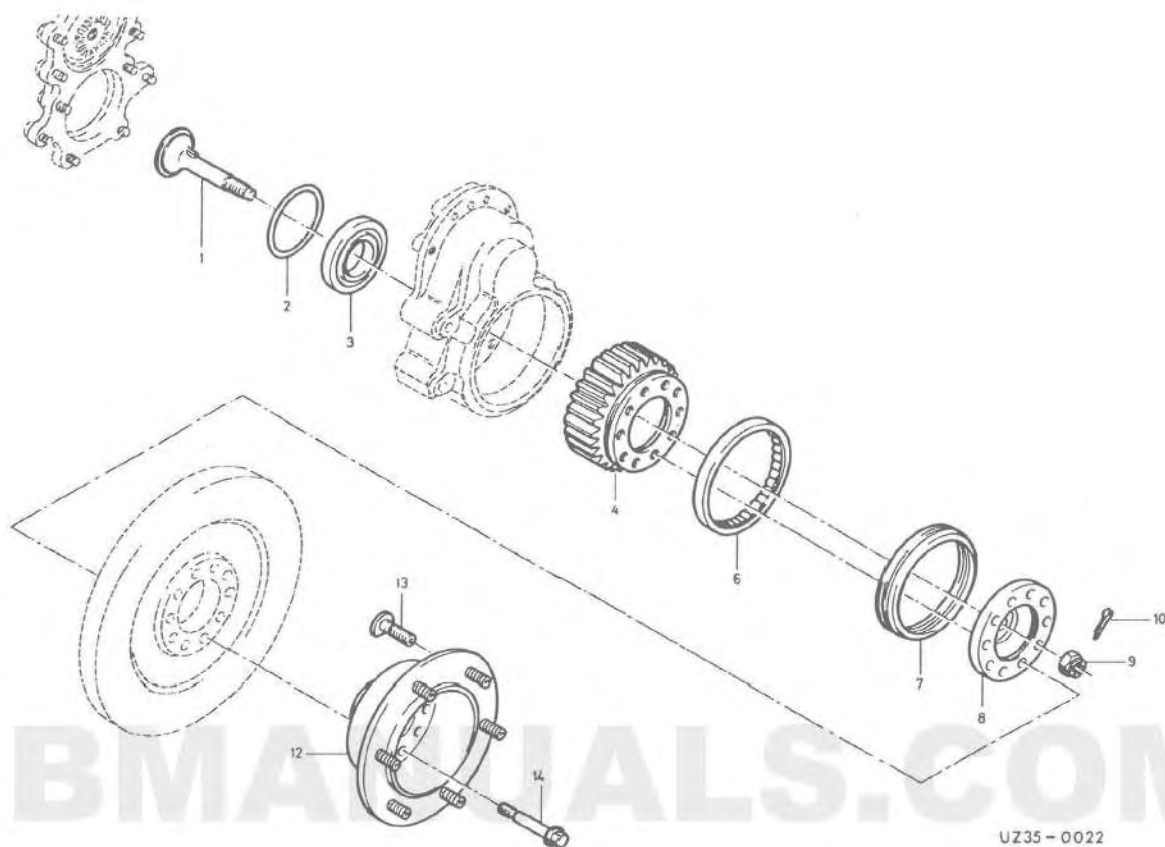


Differential

- | | |
|-----------------------------|-------------------------|
| 1 Differential housing | 18 Spring |
| 2 Differential pinion shaft | 19 Shift dog |
| 3 Differential pinion shaft | 20 Thrust washer |
| 4 Differential pinion | 21 Thrust washer |
| 5 Side gear | 22 Drive shaft |
| 6 Thrust washer | 23 Taper roller bearing |
| 7 Thrust washer | 24 Sealing ring |
| 8 Gear set | 25 Shim |
| 9 Bolt | 26 Pinion flange |
| 10 Bolt | 27 Spacing sleeve |
| 11 Nut | 28 Taper roller bearing |
| 12 Taper roller bearing | 29 Wear ring |
| 13 Sealing ring | 30 Nut |
| 14 Shim | 31 Sealing ring |
| 15 Drive shaft | 32 Retainer |
| 16 Taper roller bearing | 33 Lock washer |
| 17 Shim | |

Text and illustration do not correspond to microfiche.

Exploded view



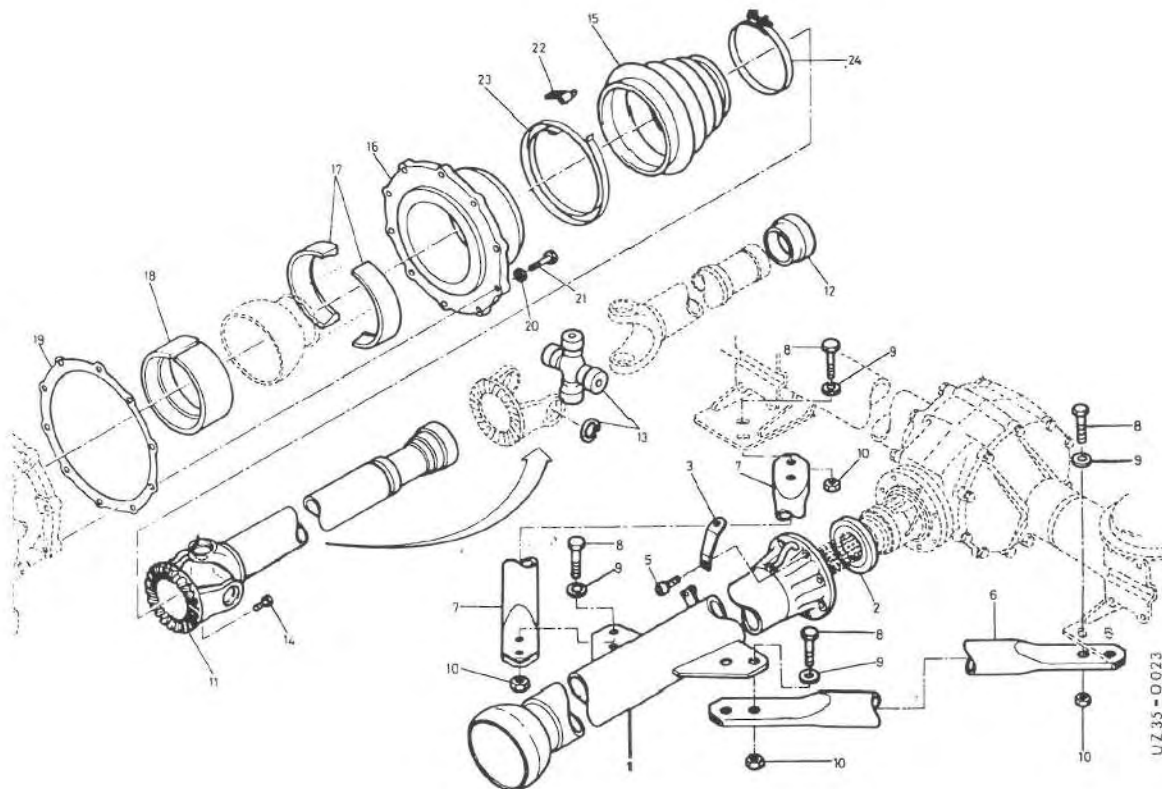
Driven gear and wheel hub

- | | |
|------------------------------|---------------|
| 1 Bolt | 8 Wear ring |
| 2 Shim | 9 Nut |
| 3 Ball bearing | 10 Split pin |
| 4 Driven gear | 12 Wheel hub |
| 6 Cylindrical roller bearing | 13 Wheel stud |
| 7 Shaft seal | 14 Bolt |

Text and illustration do not correspond to microfiche.

747.111/113

Exploded view

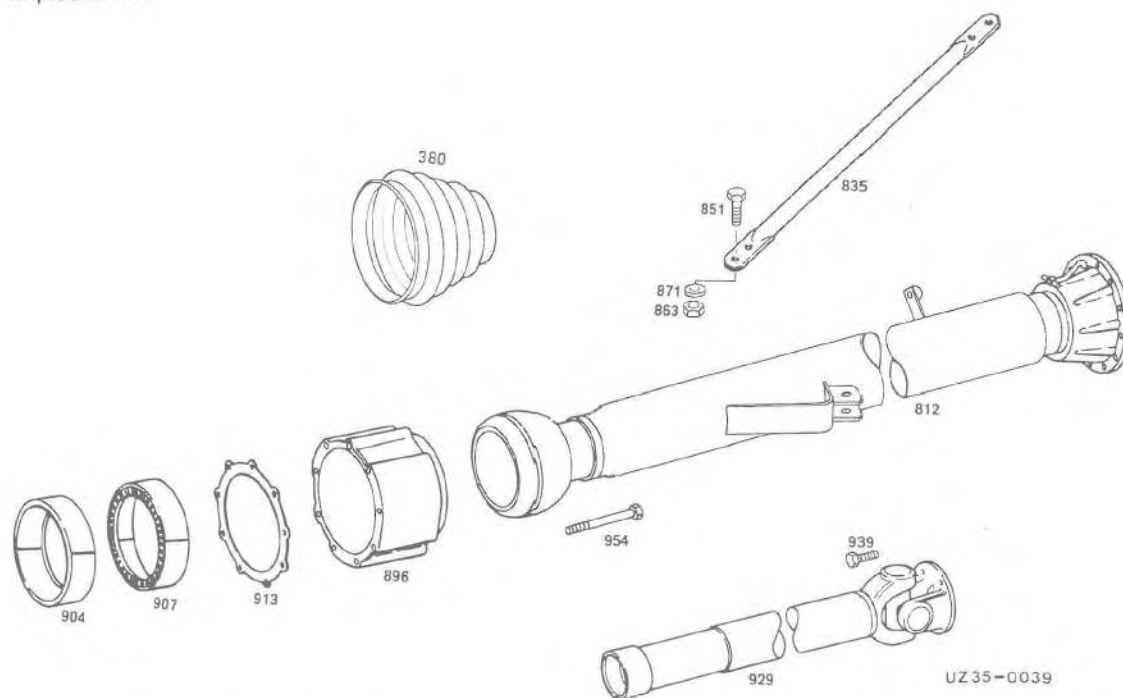


Thrust tube and propeller shaft

- | | |
|--------------------|-----------------------|
| 1 Thrust tube | 14 Bolt |
| 2 Shaft seal | 15 Rubber gaiter |
| 3 Bracket | 16 Torque ball casing |
| 5 Bolt | 17 Torque ball shell |
| 6 Axle strut | 18 Torque ball shell |
| 7 Axle strut | 19 Shim |
| 8 Bolt | 20 Washer |
| 9 Washer | 21 Bolt |
| 10 Nut | 22 Expanding clip |
| 11 Propeller shaft | 23 Strap |
| 12 Bushing | 24 Clamp |
| 13 Spider | |

Text and illustration do not correspond to microfiche.

Exploded view



Thrust tube and propeller shaft

380 Rubber sleeve
 812 Thrust tube
 835 Axle strut
 851 Bolt
 863 Nut
 871 Washer
 896 Torque ball housing

904 Torque ball shell
 907 Torque ball shell
 913 Shim
 929 Propeller shaft
 939 Bolt
 954 Bolt

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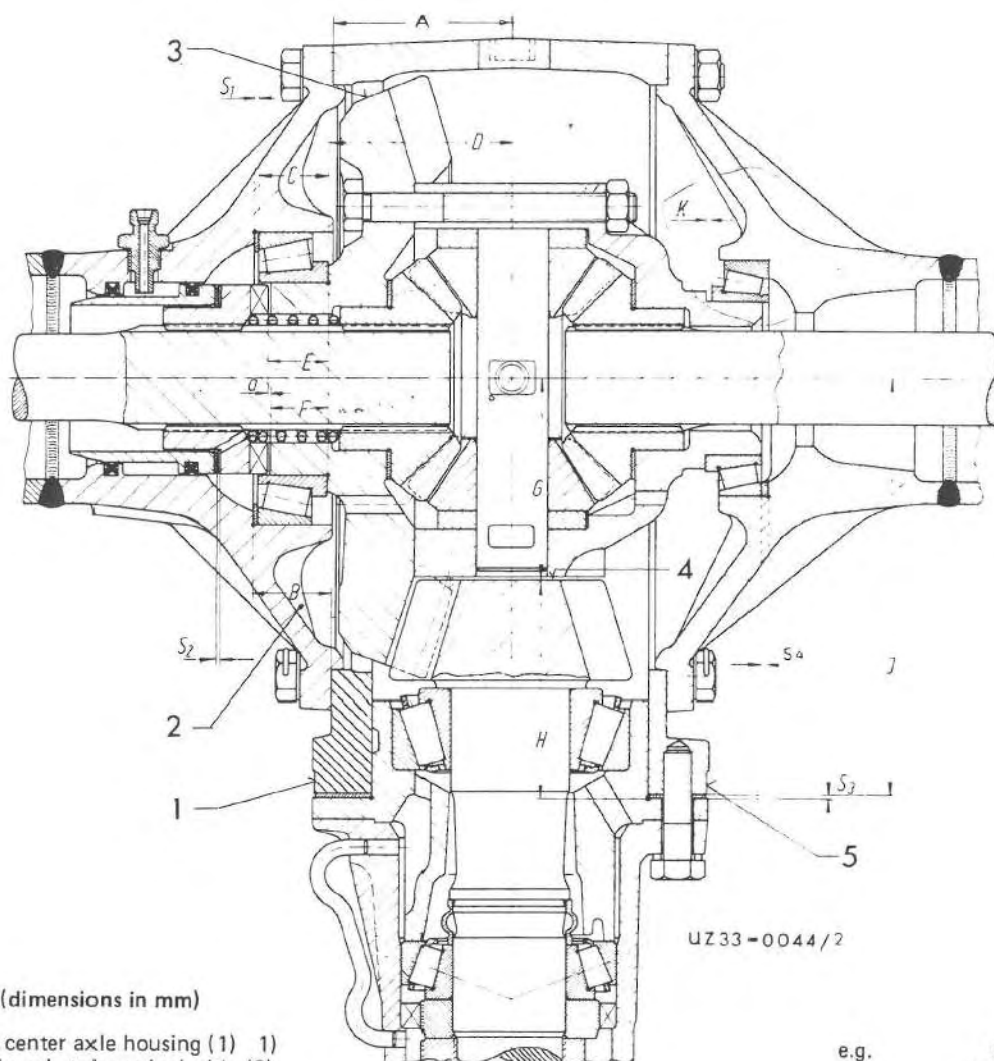
Work Sheet for Adjusting Axle Drive

Axle models 737.0-747.0/737.1-747.1

UNIMOG MB trac



Customer		Dealer	Date
Registration No.	Initial registration	Order No.	
Service hours	Chassis No.	Axle No.	
Odometer reading	Last service	carried out by	



UZ33-0044/2

Explanations (dimensions in mm)

A	etched in center axle housing (1) 1)	e.g.	90
B	stamped in axle tube at lock side (2)	e.g.	33,15
C	Overall height of large taper roller bearing, measured on a surface plate	e.g.	29,00
D	etched in ring gear (3)	e.g.	76,90
E	Measurement from collar of inner bearing race to upper edge of shift dog (washer(s) S ₁ 1, taper roller bearing, piston, thrust washer and shift dog installed in axle tube); measure with depth gage.	e.g.	34,80
F	Measurement from upper claw edge of ring gear to contact surface of bearing at ring gear	e.g.	33,30
G	Measurement from end face of drive pinion to center of differential etched in end face of drive pinion (4)	e.g.	86,15
H	Measurement from end face of drive pinion to pinion flange; measured with depth gage	e.g.	92,05
I	Measurement from center of differential to flange is stamped in top side of axle at center of axle housing (5) 2)	e.g.	77
K	Measurement from inner bearing race to contact surface of bearing at differential housing		

Important: All readings differ from axle to axle.

1) For housing width 130 mm, insert the constant 64 before the measurement (e.g. 90), and the constant 74 for housing width 140 mm.

2) Insert the constant 176 before the measurement (e.g. 77); and constant 177 for dimension 00.

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Break-away torque for adjustment of drive pinion with oiled taper roller bearings (SAE 90).

Taper roller bearing	003 981 92 05	Specified = 3,0 to 3,5 Nm 1)
	+ 004 981 28 05	
	001 981 46 05	Actual =
	+ 001 981 44 05	

Important: Rinse out taper roller bearing and insert with gear oil SAE 90.

Note readings and calculate shim thickness.

1. Shim thickness S_1 for large taper roller bearing at lock side determines backlash and contact pattern together with shim thickness S_3 .

$$S_1 = (A + B) - (C + D)$$

A =	C =	(A + B) =
+ B =	+ D =	-(C + D) =
(A + B) =	(C + D) =	$S_1 =$

2. Shim thickness S_2 between piston and differential lock determines claw play (a) of differential lock at off position.

$$S_2 = E - (F + a)$$

E =	F =	(a = claw play $0,5 \pm 0,2$ mm)
-(F+a) =	+ a =	
$S_2 =$	F + a =	

3. Shim thickness S_3 at flange of drive pinion bearing determines amount by which drive pinion engages ring gear.

$$S_3 = (G + H) - J$$

G =	(G + H) =
+ H =	- J =
(G + H) =	$S_3 =$

4. Shim thickness S_4 at small taper roller bearing for front axle

$$S_4 = K$$

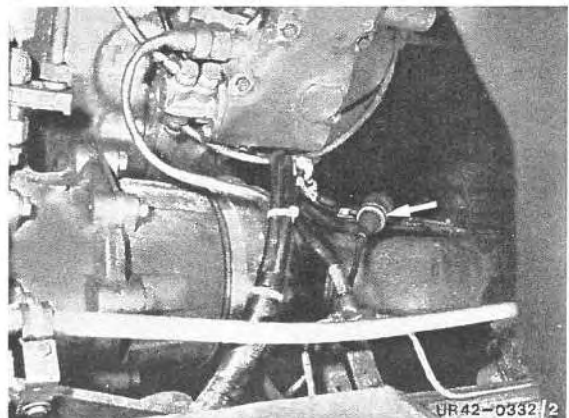
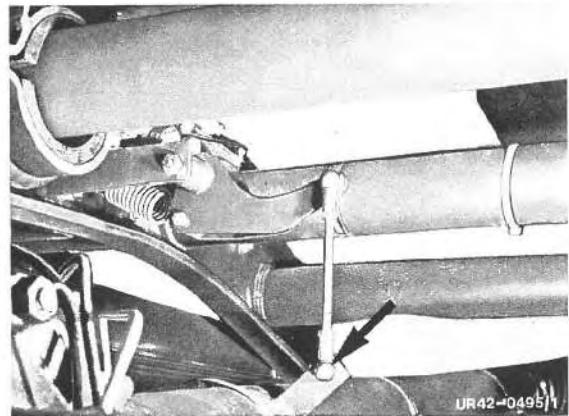
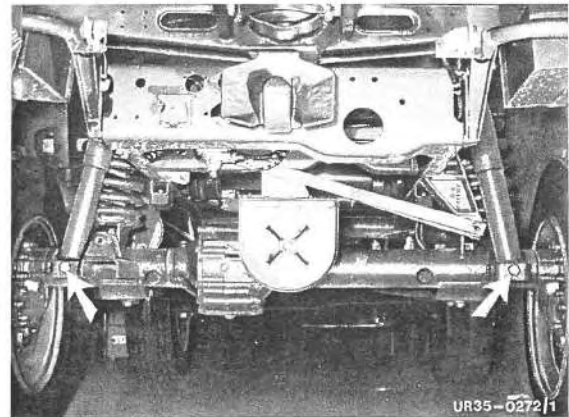
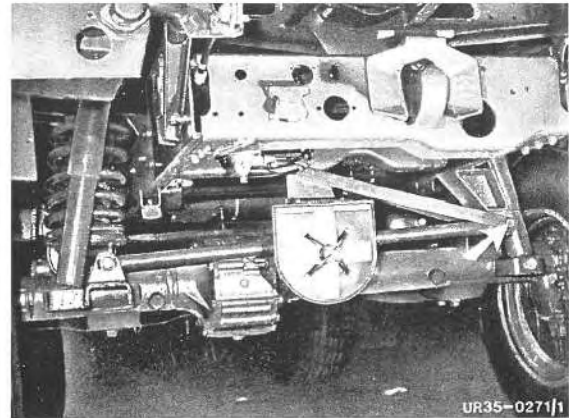
Explanations to shim thickness S_4 The coil spring of the differential lock between the shift dog and ring gear is **not** to be fitted for measurement.

- 4.1 Force outer bearing race into axle tube without shims.
- 4.2 Force taper roller bearing with inner race onto bearing seat of differential housing by half the bearing thickness.
- 4.3 Introduce differential into center axle housing at lock side.
- 4.4 Position axle tube on center axle housing and torque **four opposite bolts** to 138 Nm 2) while steadily turning ring gear.
- 4.5 After tightening bolts, continue to rotate ring gear while knocking (plastic-headed hammer) flange part of axle tube to eliminate stresses from bearing seat.
- 4.6 Remove axle tube and withdraw outer bearing race again.
- 4.7 Measure gap K with feeler gage, insert shims S_4 of calculated thickness beneath outer race of taper roller bearing, and force race into position.
- 4.8 Take differential out.
- 4.9 Fully press inner race of taper roller bearing onto bearing seat of differential housing.

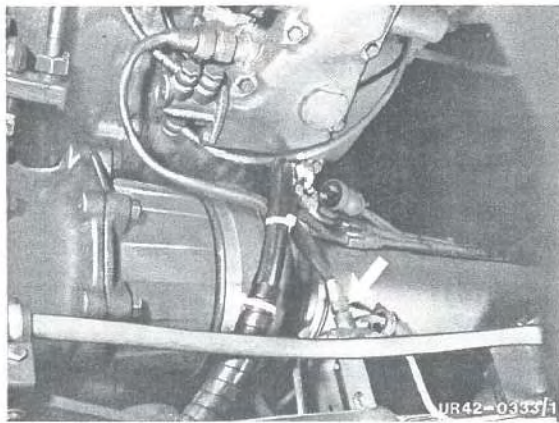
- 1) Taper roller bearing 001 981 44 05 Specified = 2,0 to 2,2 Nm
- 2) For axle model 373/747.111, torque to 200 Nm.

Removal

- 1 Remove spare wheel.
- 2 Unscrew transverse link at frame .
- 3 Unscrew shock absorber at axle.
- 4 Release and remove pull rod of ALB modulator.
- 5 Unscrew line for brake lining wear indicator.

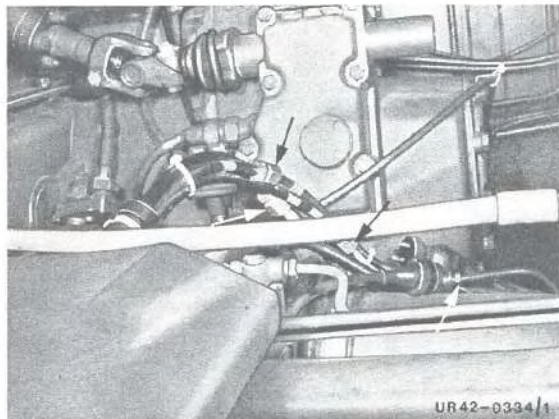


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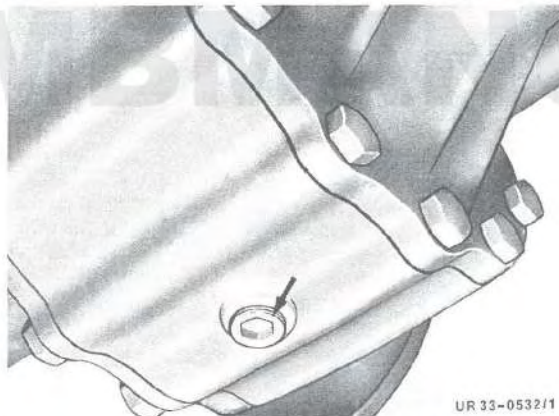


6 Unscrew brake line.

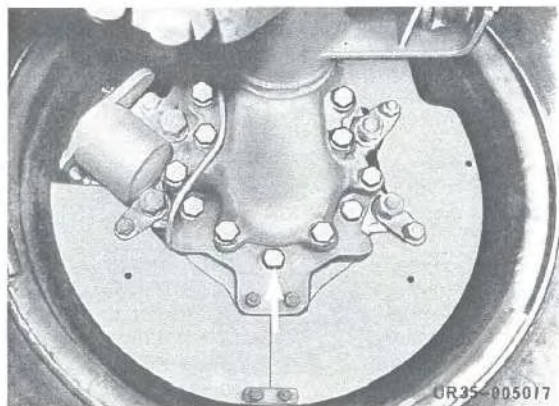
Note: Seal end of brake line.



7 Unscrew lines to differential lock and spring-hausting lines to axle and spring-loaded brake cylinders.

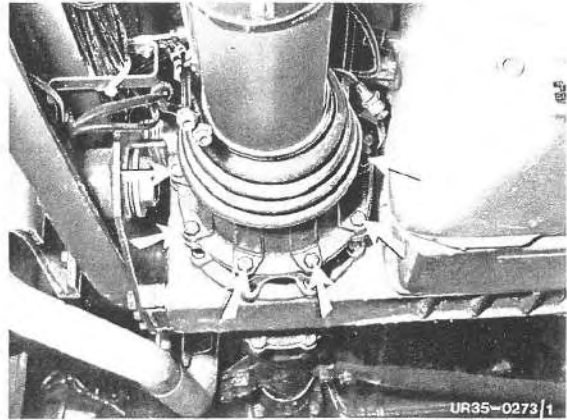


8 Drain oil off center axle housing.



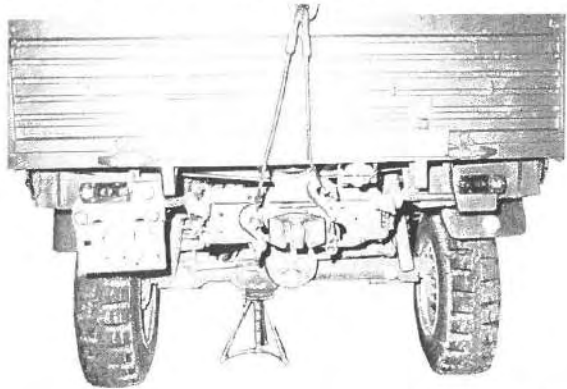
9 Drain oil off wheel hub drive.

- 10 Unscrew torque ball casing at transmission.

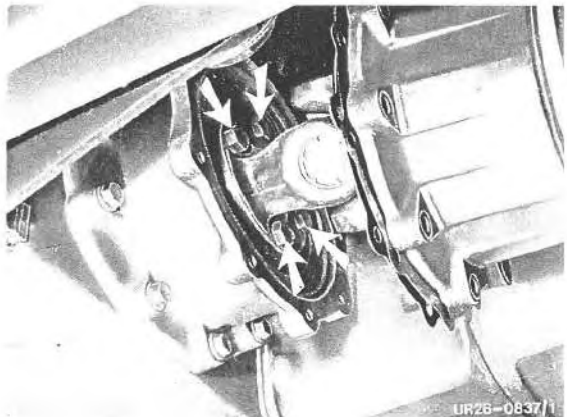


- 11 Using lifting gear, raise vehicle off springs, release spring-loaded cylinder, draw axle to rear and support torque tube.

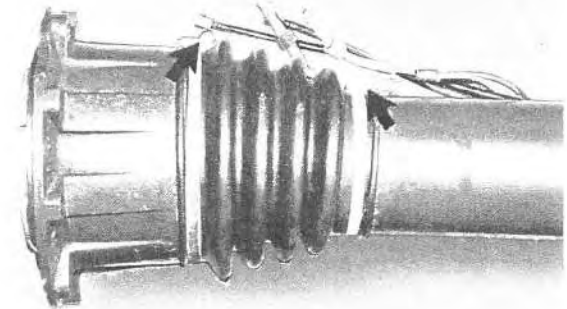
Note: To avoid damage to planing, place timber between chain and planking.



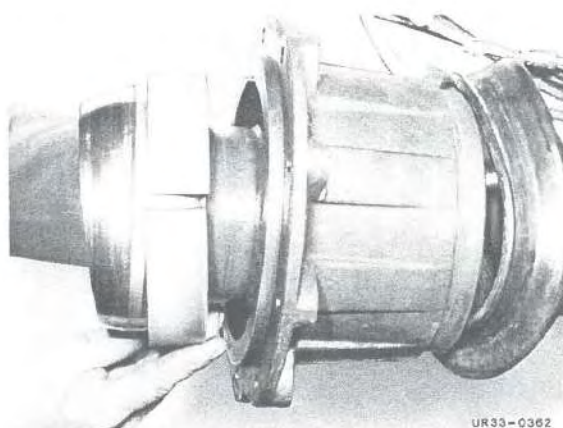
- 12 Remove front torque ball shell halves.
 13 Detach propeller shaft at transmission.
 14 Draw rear axle out to rear.
 15 Lower vehicle onto trestles.



- 16 Detach clamps and draw rubber gaiter back.

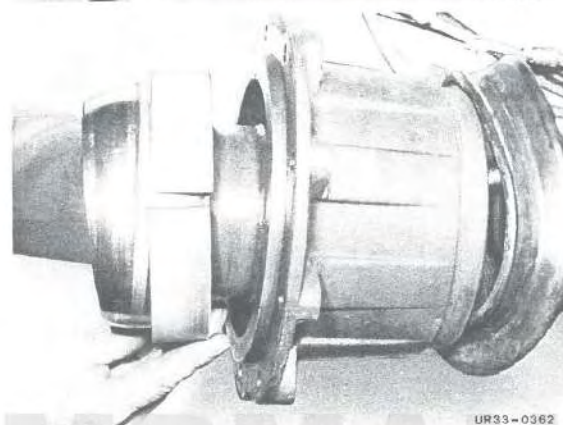


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17 Draw torque ball casing back and remove rear torque ball shell halves.

18 Clean and check torque ball shell halves and torque ball casing, exchange if necessary.



Preparation for installation

1 Carefully deburr torque ball casing and torque ball shell halves.

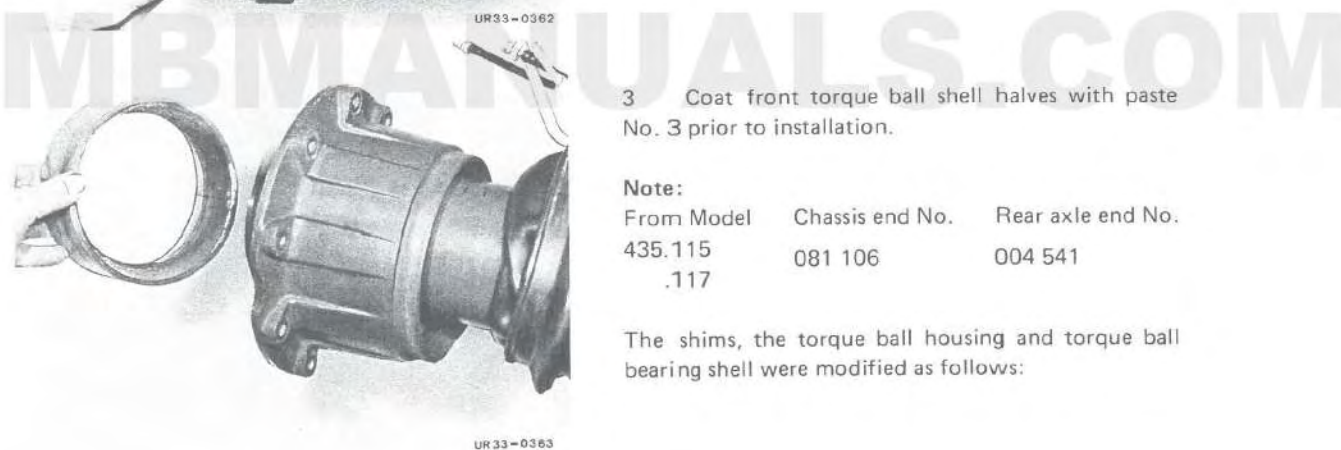
2 Coat rear torque ball shell halves with paste No. 3 prior to installation.

3 Coat front torque ball shell halves with paste No. 3 prior to installation.

Note:

From Model	Chassis end No.	Rear axle end No.
435.115	081 106	004 541
.117		

The shims, the torque ball housing and torque ball bearing shell were modified as follows:



Item	Designation	Modification
1	Torque ball housing	Collar dia. 195 mm shortened from 8 mm to 5 mm (see ill., Item a and b resp.)
2	Torque ball bearing shell front	Overall length shortened from 66.5 mm to 62 mm
3	Change in shape	see ill.
4	Sealing compound (Terostat 56)	Used as seal between torque ball housing and transmission flange

4 Attach special tool No. 14 and **evenly** tighten fastening bolts until a manual effort of 200 to 300 N is reached at the extension tube.

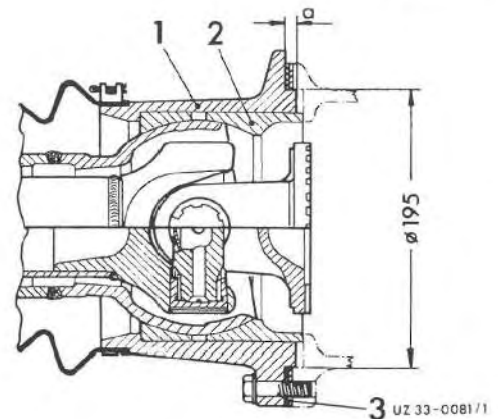
Note: Remove shims if necessary manual effort is inadequate, and add shims if manual effort is excessive. When using the new shims it is necessary to insert the shims if the manual effort is inadequate and to remove shims if the manual effort is excessive.



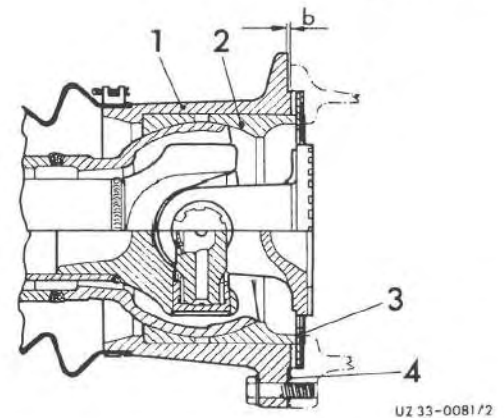
UR 33-0553

5 Previous position of shims (3)
a = 8 mm

Note: The sealing ring is not installed in the models SA 35 973 and 35 974 (front and rear axle sections capable of fording).



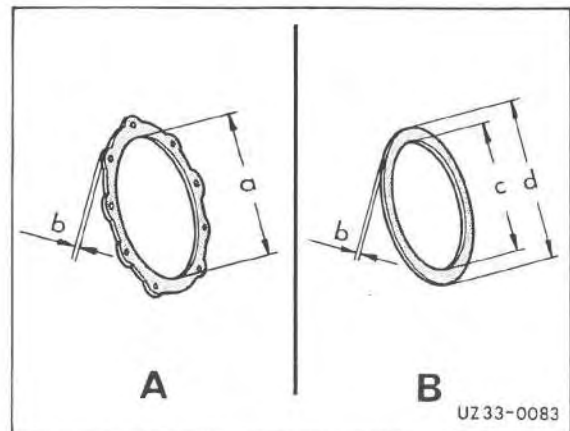
6 New position of shims (3)
b = 5 mm



7 Install shims for measuring purposes.

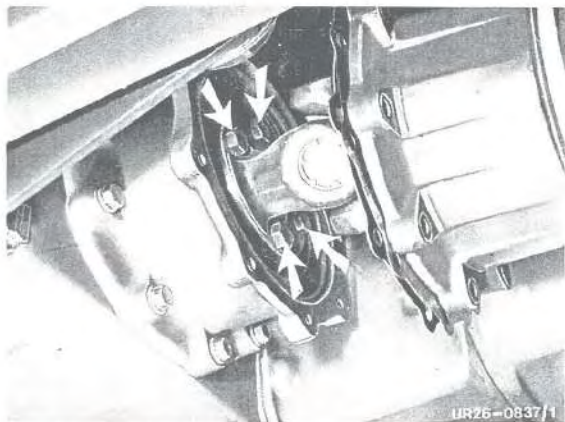
Note:

- A Previous shims
- B New shims
- a $\varnothing$ 205 mm
- b 0,2; 0,5; 1,0 mm
- c $\varnothing$ 148 mm
- d $\varnothing$ 194 mm



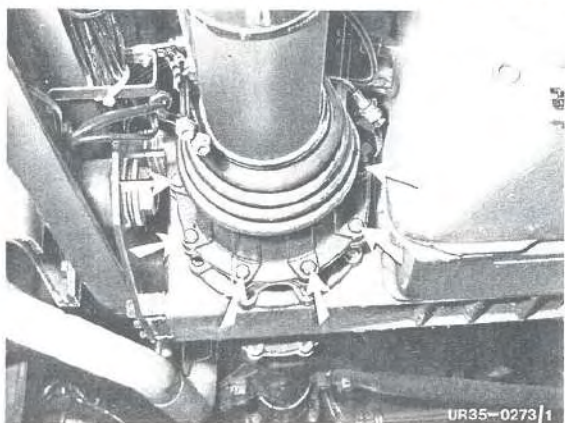
35.6 Removal and Installation of Rear Axle

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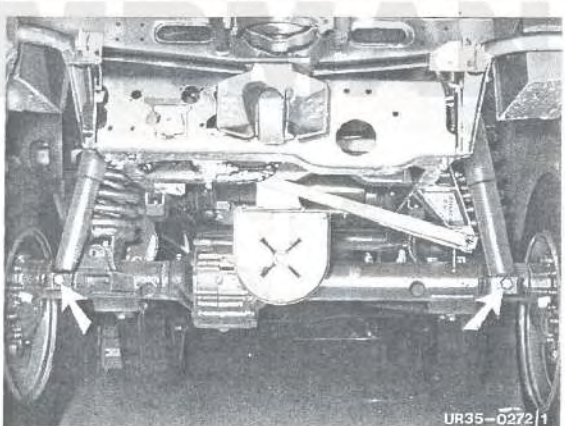


Installation

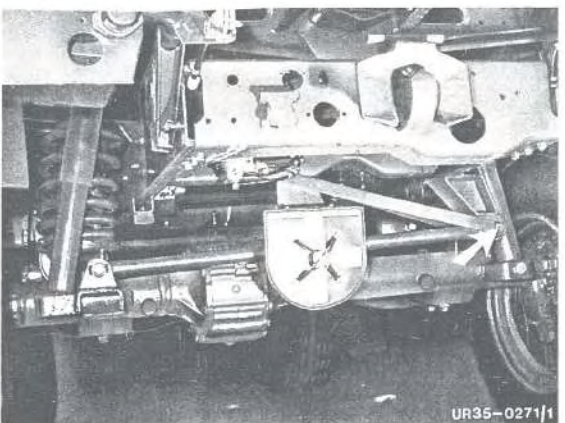
- 1 Raise vehicle and move axle into position.
- 2 Attach drive shaft to transmission.



- 3 Attach torque ball casing together with the shims of calculated thickness and tighten.
Tightening torques, see 1.5/1.

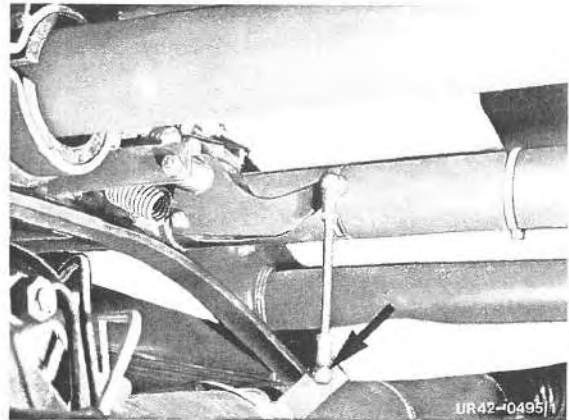


- 4 Install shock absorber.
Tightening torques, see 1.5/1.

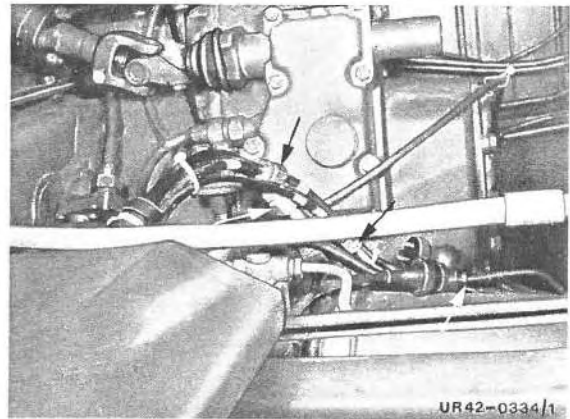


- 5 Attach transverse link to frame.
Tightening torques, see 1.5/1.

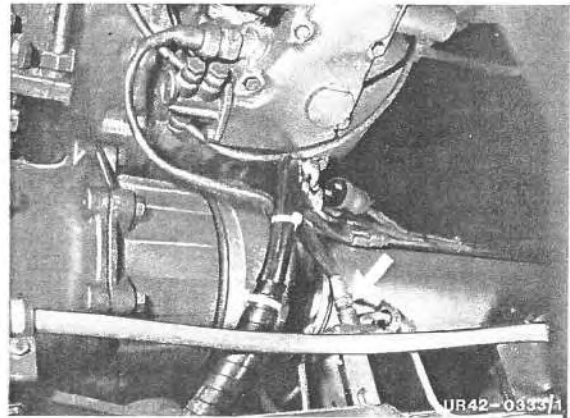
6 Attach and secure pull rod of ALB modulator.
See 42.11—5.1/1 or 42.11—5.2/1 for checking adjustment of ALB modulator.



7 Attach lines to differential lock and spring-loaded cylinders. Also fit air supply and exhausting lines to axle and spring-loaded brake cylinders.



8 Attach brake line.



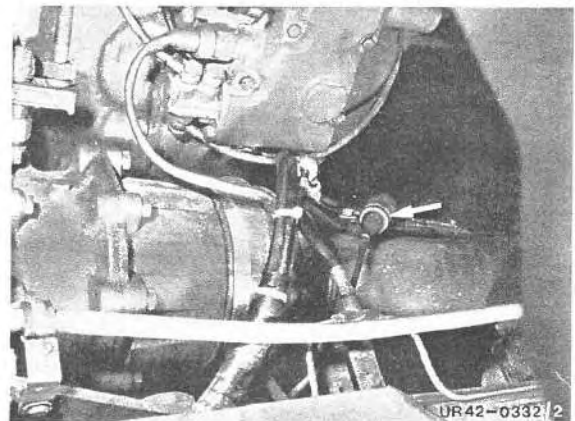
9 Attach line for brake lining wear indicator.

10 Fill center axle housing with oil, referring to 1.1/2 for quantity.

11 Fill wheel hub drive with oil, referring to 1.1/2 for quantity.

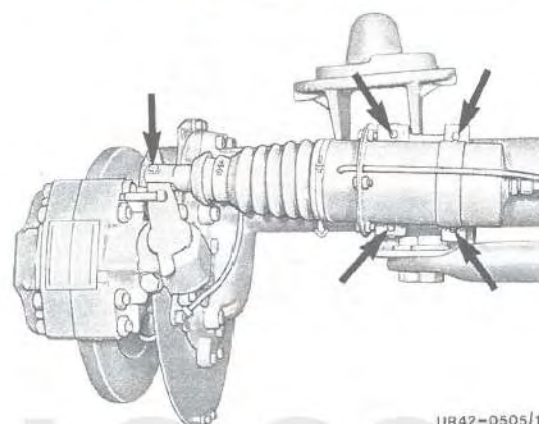
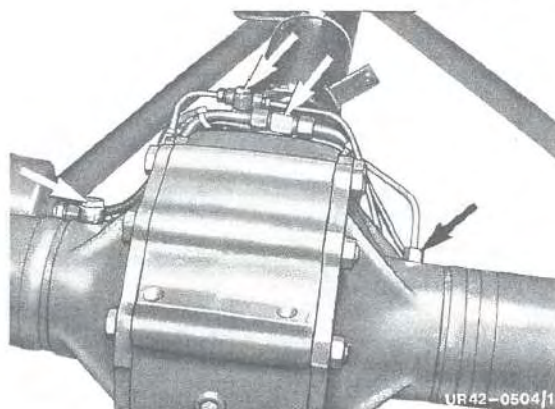
12 Bleed brake system referring to 42.11—6.1/1.

13 Check brake system for leakage.

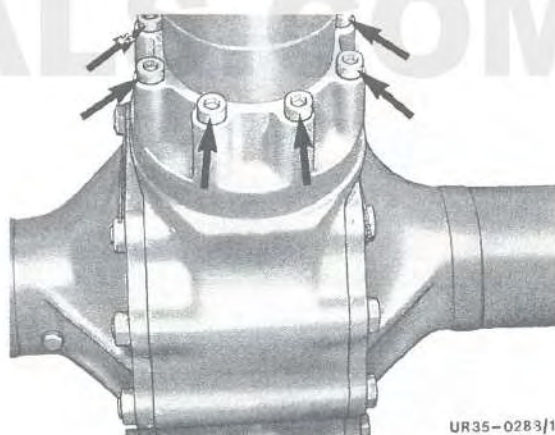


Disassembly

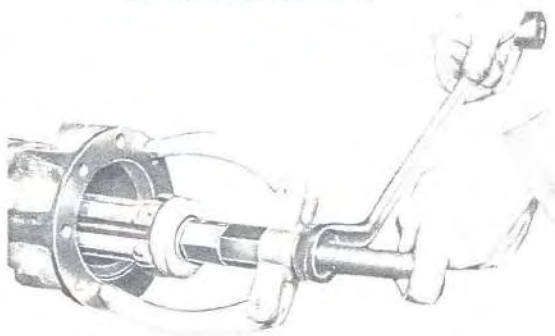
- 1 Remove rear axle, referring to 2.1/1.
- 2 Clean rear axle, clamp in trestle and detach wheel.
- 3 Detach brake lines, line for differential lock and axle ventilation, as well as line for spring-loaded cylinder.
- 4 Detach spring-loaded cylinder.
- 5 Detach axle struts.



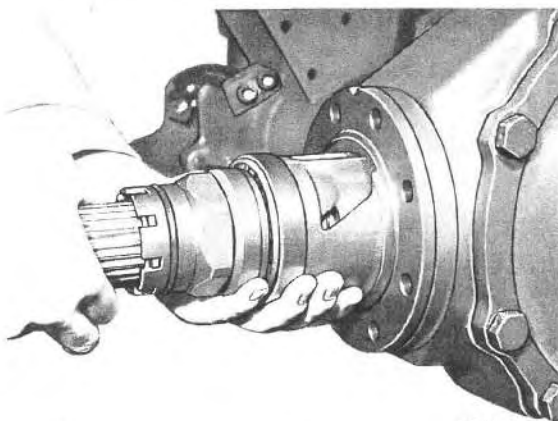
- 6 Using special tool No. 2, unscrew torque tube and detach propeller shaft.
- 7 Detach wheel hub drive, referring to 6.1/1.



- 8 Check shaft seal in torque tube for wear, extracting with puller and special tool No. 9 if necessary.



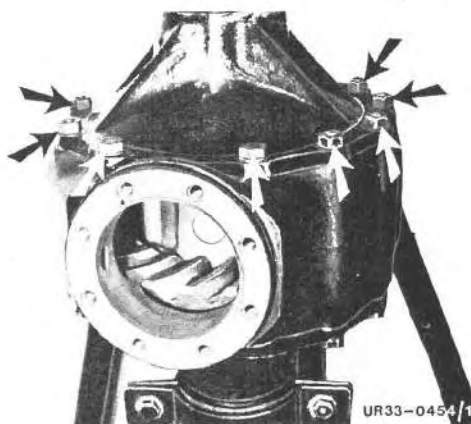
747.111/113



UR33-0562

- 9 Remove drive pinion bearing.

Note: Watch out for shims between pinion flange and axle housing.



UR33-0454/1

- 10 Turn axle by 90°, with long axle tube pointing downward.

- 11 Unscrew axle tube at center axle housing.

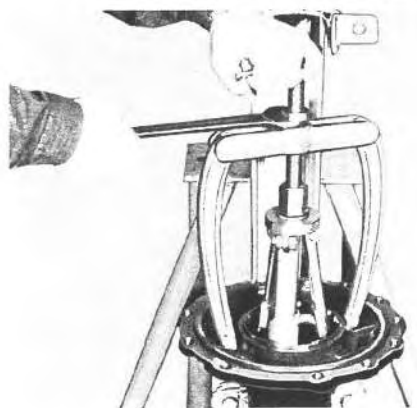


UR33-0455

- 12 Take differential out of center axle housing.

- 13 Remove center axle housing.

- 14 Take release spring, shift dog, shims and piston out of long axle tube.



UR33-0026

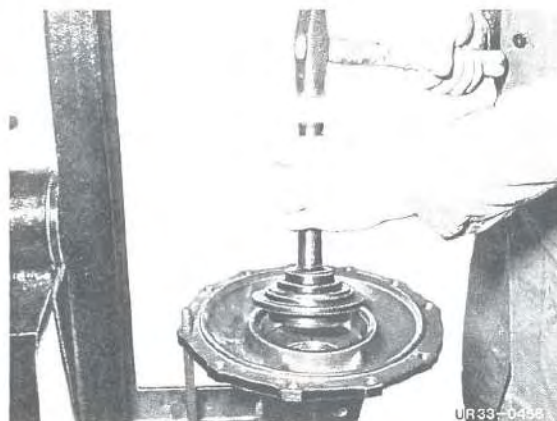
- 15 Using puller and special tool No. 9, draw outer races of taper roller bearings off both axle tubes.

Note: Remove Nilos ring from short axle tube.

- 16 Clean and check all parts, exchange if necessary.

Assembly

Note: Prior to assembling axle, adjust axle drive with reference to „Work sheet for adjusting axle drive“. During the operation „Adjustment of axle drive“ the axle tube at lock side has already been clamped in trestle, and outer taper roller bearing race has been forced into position with shims, using special tool No. 21.



- 1 Force shims and outer taper roller bearing race into short axle tube, using special tool No. 21.

Note:

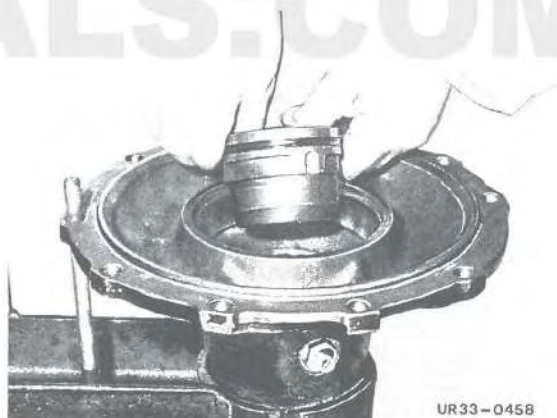
From Model	Chassis end No.	Rear axle No.
435.111	070 262	003 566

Nilos ring isn't installed anymore.



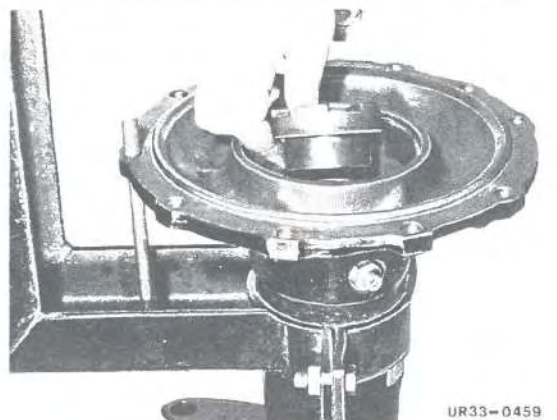
- 2 Coat differential lock piston with anti-corrosion oil No. 6 and push fully home into long axle tube with sealing rings.

Note: Make sure that adapter socket is seated correctly relative to guide slot in piston.

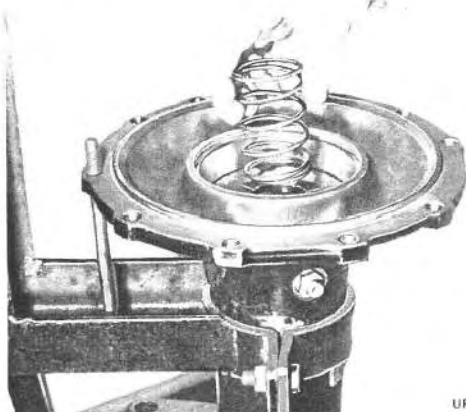


- 3 Fit shift dog with thrust washers and shims in piston.

- 4 Check differential lock for leakage, referring to 7.1/1.

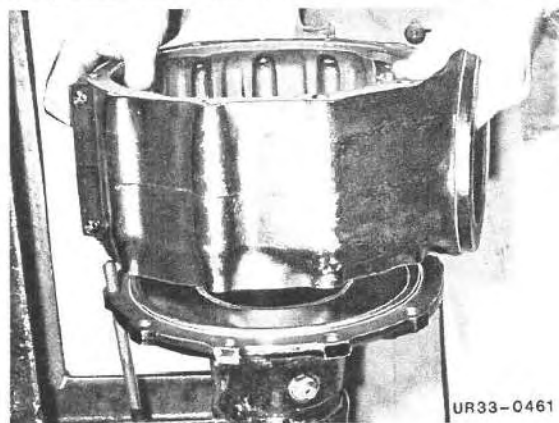


747.111/113



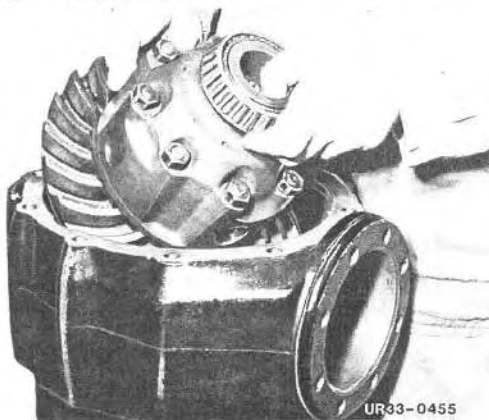
UR33-0460

- 5 Introduce release spring into shift dog.



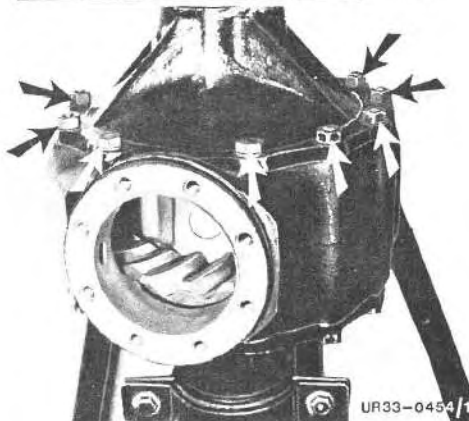
UR33-0461

- 6 Coat contact surface of long axle tube with sealant No. 8 and place center axle housing in position.



UR33-0455

- 7 Install differential.



UR33-0454/1

- 8 Coat contact surface of short axle tube with sealant No. 8. Torque axle tube and coat bolts with sealant No. 1.

See 1.5/1 for tightening torques.

Note: Secure short bolts with wire.

- 9 Turn rear axle by 90° in trestle.